

Isolation basin stratigraphy and Holocene relative sea-level change on Arveprinsen Ejland, Disko Bugt, West Greenland

ANTONY J. LONG¹*, DAVID H. ROBERTS² and MATTHEW R. WRIGHT¹

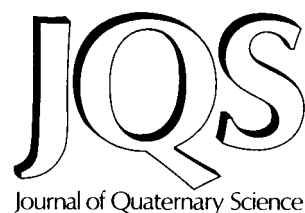
¹Environmental Research Centre, Department of Geography, University of Durham, Science Site, South Road, Durham DH1 3LE, England

²Cartref, Tafarn-y-Gelyn, Ruthin Road, Llanferres, Mold, CH7 5SQ, Wales

Long, A. J., Roberts, D. H. and Wright, M. R. 1999. Isolation basin stratigraphy and Holocene relative sea-level change on Arveprinsen Ejland, Disko Bugt, West Greenland. *J. Quaternary Sci.*, Vol. 14, pp. 323–345, ISSN 0267–8179.

Received 27 July 1998; revised 18 November 1998; accepted 19 November 1998

ABSTRACT: This paper presents the results of an investigation into Holocene relative sea-level (RSL) change, isostatic rebound and ice sheet dynamics in Disko Bugt, West Greenland. Data collected from nine isolation basins on Arveprinsen Ejland, east Disko Bugt, show that mean sea level fell continuously from ca. 70 m at 9.9 ka cal. yr BP (8.9 ka ¹⁴C yr BP) to reach a minimum of ca. –5 m at 2.8 ka cal. yr BP (2.5 ka ¹⁴C yr BP), before rising to the present day. A west–east gradient in isostatic uplift across Disko Bugt is confirmed, with reduced rebound observed in east Disko Bugt. However, RSL differences (up to 20 m at 7.8 ka to 6.8 ka cal. yr BP (7 ka to 6 ka ¹⁴C yr BP)) also exist within east Disko Bugt, suggesting a significant north–south component to the area’s isostatic history. The observed magnitude and timing of late Holocene RSL rise is not compatible with regional forebulge collapse. Instead, RSL rise began first in the eastern part of the bay, as might be expected under a scenario of crustal subsidence caused by neoglacial ice sheet readvance. The results of this study demonstrate the potential of isolation basin data for local and regional RSL studies in Greenland, and the importance of avoiding data compilations from areas where the isobase orientation is uncertain. Copyright © 1999 John Wiley & Sons, Ltd.



KEYWORDS: Greenland Ice Sheet; sea level; isostasy; diatoms; neoglacial.

Introduction

For over 30 yr established approaches to relative sea-level (RSL) studies in Greenland have relied on chronologies developed primarily from radiocarbon dated marine shells found on or within raised beaches and deltas (e.g. Weidick, 1972, 1976; Donner and Junger, 1975; Kelly, 1985; Funder, 1989; Ingolfsson *et al.*, 1990). Although supplemented with other dates from bone, driftwood, lake sediment and archaeological sources, uncertainty surrounds some of these data and their associated RSL reconstructions. For example, shells may be reworked and/or live at an uncertain water depth, thus providing only lower limiting data for RSL reconstructions. In a careful review of existing RSL records from Greenland, Kelly (1985, p. 467) highlighted these concerns by concluding that ‘in no area do the data really justify expressing the emergence by a single well-defined curve;

although some of the uncertainty may arise from the need to include data from over relatively large areas, most of it is inherent in the data themselves’.

Because of these problems, and because of the close relationship that exists between chronologies of sea-level change and ice sheet reconstructions (e.g. Weidick, 1993), improving the quality of existing RSL records is clearly an important research priority. Recent work within Disko Bugt, West Greenland, has sought to rectify some of the data deficiencies through an emphasis on the morpho-sedimentary evidence for RSL change (e.g. Rasch *et al.*, 1997; Rasch and Jensen, 1997) as well as by attempting smaller areal compilations of data (Rasch and Nielsen, 1994, 1995). However, interregional as oppose to intersite comparisons remain the norm in Greenland RSL studies.

This paper presents one of the first local RSL records that spans the full post-glacial period from West Greenland. The study site is on Arveprinsen Ejland, located in east Disko Bugt, and the work relies on the analysis of sediments preserved in a staircase of nine isolation basins, which occur at elevations from below present mean sea level (MSL) to the local marine limit. Although isolation basins have been studied previously in Greenland (e.g. Foged, 1972, 1973, 1977; Bennike, 1995), this work is novel in the number and

*Correspondence to: A. J. Long, Department of Geography, University of Durham, Science Site, South Road, Durham DH1 3LE, England.
E-mail: A. J. Long@Durham.ac.uk

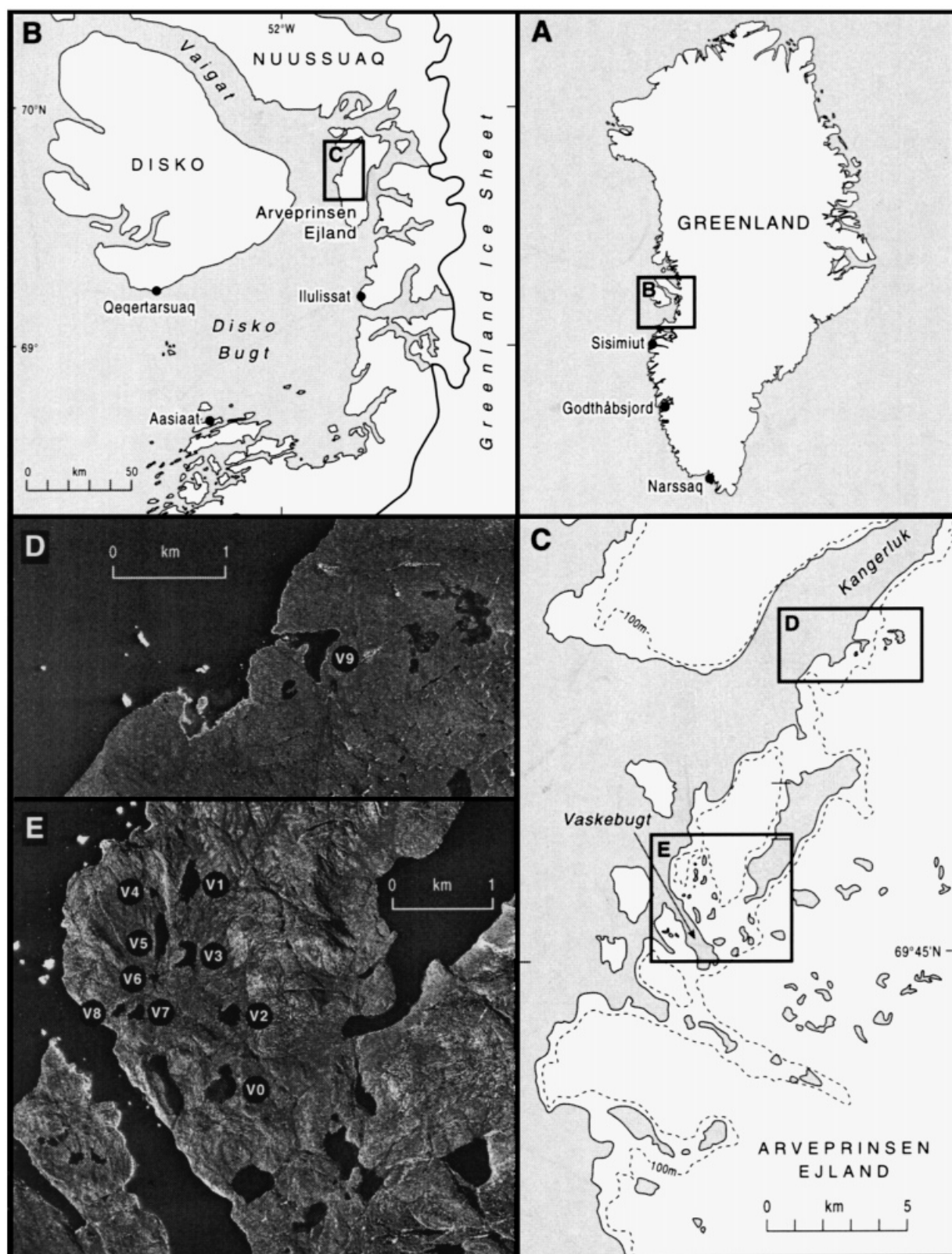


Figure 1 Location map showing Greenland and the position of Disko Bugt (A), Disko Bugt and the location of Arveprinsen Ejland (B), the field area on the west coast of Arveprinsen Ejland (C), and the locations of the lake basins studied (D and E). Basins are numbered from V0 (the highest) to V9 (the lowest). The air photographs in Figure 1d and 1e are copyright: Kort & Matrikelstyrelsen, Denmark (A. 229–98).

density of the sites investigated. Comparison of this work with other data collected previously from Disko Bugt (Weidick, 1993, 1996; Rasch, 1997; Rasch and Jensen, 1997) enable preliminary testing of models of isostatic rebound in the study area.

The study area

Disko Bugt is a large embayment on the central west coast of Greenland between 68°30'N and 69°15'N and 50°00'W and 54°00'W (Fig. 1a and b). The bedrock geology of Disko, which occupies the centre of the bay, consists of relatively soft Tertiary basalts, whereas much of the surrounding mainland (and Arveprinsen Ejland) is underlain by Precambrian gneisses (Escher *et al.*, 1976). Water depths within the Disko Bugt embayment are typically between 200 m and 600 m, but in fjords these may increase to over 1000 m. A southerly extension of a gneiss ridge that crosses Disko defines the western limit of this embayment along a north–south line linking Qeqertarsuaq and Aasiaat (Fig. 1b).

The climate of Disko Bugt is strongly influenced by the relatively warm waters of the northward flowing West Greenland current (Funder, 1989). The mean July air temperature is about 7°C and the mean annual air temperature is –4°C (Nielsen *et al.*, 1995; Hansen *et al.*, 1997). Disko Bugt is ice free between mid-April and mid-January.

Arveprinsen Ejland lies in eastern Disko Bugt, is ca. 30 × 15 km in size and has an ice free interior which rises to ca. 800 m above sea level. The surface relief of the island is undulating and heavily ice scoured, with abundant outcrops of bedrock and limited surficial sediment cover (e.g. Fig. 1e). The Vaskebugt area was chosen for analysis, a small west-facing fjord where lake basins below the 100 m contour abound (Fig. 1c–e), providing the opportunity for a detailed local study of RSL change.

Previous work

At the culmination of the Weichselian glaciation (the Sisimiut Stade) the Greenland Ice Sheet extended to the outer coast of the mainland, with a large glacier occupying Disko Bugt and an ice shelf probably extending into Baffin Bay (Bennike *et al.*, 1994). Initial break-up of the marine-based portion of the ice sheet occurred sometime between 16.8 ka and 11.1 ka cal. yr BP (14 ka and 10 ka ¹⁴C yr BP) (Funder and Hansen, 1996), triggered by rapidly rising eustatic sea level. Funder and Hansen (1996) draw comparisons between this initial deglaciation and the destabilising and rapid collapse of other Northern Hemisphere marine-based ice sheets at this time, such as the Barents Sea (e.g. Elverhøi *et al.*, 1993). This first phase of deglaciation was followed by a more gradual retreat of the dry (land-based) ice during the early Holocene (Weidick, 1985; Funder and Hansen, 1996).

Maps depicting early Holocene ice recession in Disko Bugt are contradictory. Some, such as that proposed by Weidick (1996), position the 11.1 ka cal. yr BP (10 ka ¹⁴C yr BP) ice sheet margin close to the bedrock high between Aasiaat and Qeqertarsuaq referred to above (Fig. 1b). Others, including Funder and Hansen (1996), suggest a more easterly position, arguing that much of Disko Bugt was ice-free by this time (if not before). There is closer agreement that by

8.8 ka to 7.8 ka cal. yr BP (8 ka to 7 ka ¹⁴C yr BP) the ice margin had retreated from the outer coast to the bay's eastern shores (Funder, 1989; Weidick *et al.*, 1990; Weidick, 1993). A temporary early Holocene local glacier readvance on Disko, the Disko Stade, interrupted this retreat but was complete by ca. 10 ka cal. yr BP (9 ka ¹⁴C yr BP) (Kelly, 1985; Ingolfsson *et al.*, 1990). No evidence of the Disko Stade is recorded on the mainland and this readvance is thought to reflect a local change in upper-air wind conditions and, an accompanying shift in equilibrium line altitude (Ingolfsson *et al.*, 1990).

During the climatic optimum of the mid-Holocene the ice sheet contracted further to attain a position by 6.8 ka cal. yr BP (6 ka ¹⁴C yr BP) some 15 km to 20 km behind the present ice sheet margin (Kelly, 1980; Weidick, 1985; Funder, 1989). Using the Camp Century isotope record (Dansgaard *et al.*, 1984), Weidick *et al.*, (1990) suggest that average annual temperatures at Jakobshavn were 2°C warmer than present during this period, with a 10% to 15% increase in the duration of the ablation season at the ice margin. These estimates are similar to those reported recently by Dahl-Jensen *et al.* (1998) based on temperature profiles from the GRIP and Dye 3 ice-cores, which indicate temperatures ca. 2.5°C above present between 8 ka and 5 ka BP. The ice-free zone of West Greenland may have been 15% greater at this time (Weidick *et al.*, 1990).

The Greenland Ice Sheet subsequently readvanced as climate cooled during the early neoglacal, from about 4.4 ka cal. yr BP (4 ka ¹⁴C yr BP) onwards. This readvance, which probably had several pulses, culminated in deposition of the 'historic' moraines that formed during the Little Ice Age. Kelly (1980) notes that evidence relating to the early neoglacal ice sheet readvance (i.e. prior to the Medieval Warm Period) is limited, probably due to sediment reworking by the late neoglacal readvance of the Little Ice Age. Down-core temperature profiles from the GRIP and the Dye 3 boreholes (Dahl-Jensen *et al.*, 1998) suggest temperatures falling during the early neoglacal to ca. 0.5°C below present values at 2 ka BP. Medieval Warm Period temperatures (about 1 ka BP) were ca. 1°C warmer than present, and the Little Ice Age had two minimums at ca. 0.5 ka and 0.1 ka BP.

Patterns of Holocene RSL change in Disko Bugt agree broadly with the timing of ice loading and unloading described above. From the marine limit, RSL fell continuously in all areas until ca. 4.4 ka to 3.2 ka cal. yr BP (4 ka to 3 ka ¹⁴C yr BP), after which a switch from emergence to submergence occurred (Rasch and Nielsen, 1994, 1995; Rasch and Jensen, 1997; Rasch *et al.*, 1997). Maximum RSL fall is recorded in the vicinity of Aasiaat (ca. 120 m) (Fig. 2), with reduced values (< 100 m) in those areas that either escaped significant ice load during the Weichselian (e.g. northwest Disko) or were comparatively recently deglaciated, such as those close to the present ice sheet (Donner, 1978; Ingolfsson *et al.*, 1990; Rasch, in press). A late Holocene switch from emergence to submergence is observed throughout West Greenland. Kelly (1985) termed the accompanying rise in RSL the 'Sandnes transgression' after the Norse settlement at Sandnes, Godthåbsfjord, which has been lowered by as much as 5 m to 6 m since 0.6 ka cal. yr BP. Other evidence for submergence includes subaerially weathered surface layers of marine clays, which now lie 5 m and 8 m below present sea level at Sisimiut and Narssaq respectively (Fig. 1a) (Kelly, 1980). A dessication horizon at the latter has been radiocarbon dated to 3080 ± 110 yr BP (Ua-2425, 3477–2954 cal. yr BP (Weidick, 1993)).

Two models exist to explain the late Holocene RSL rise in West Greenland. Firstly, Rasch and Nielsen (1995) argue

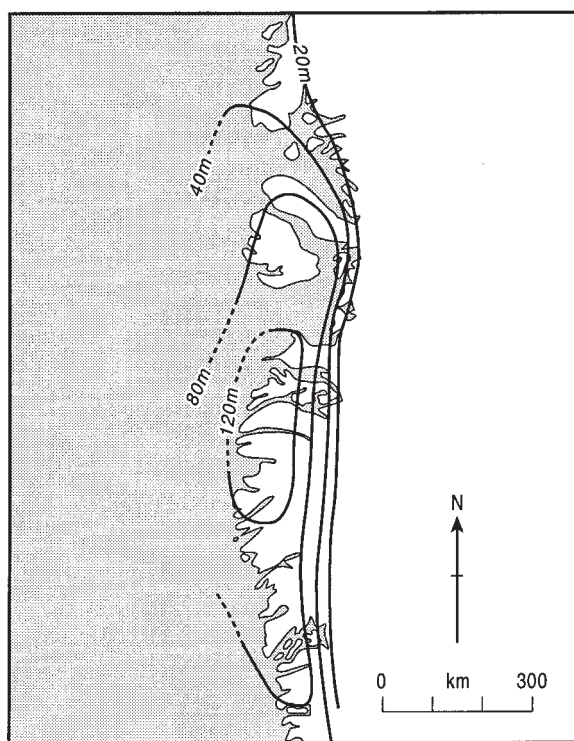


Figure 2 Map depicting the elevation (in metres above sea level) of the marine limit in central West Greenland (from Weidick, 1992). Note the doming of the marine limit over the ice-free coastal land to the south of Disko Bugt.

that the J-shaped RSL curves which they and others propose (e.g. Donner and Junger, 1975) reflect the regional migration of a collapsing forebulge associated with the Greenland Ice Sheet. A second explanation for this subsidence is provided by Kelly (1980) and Weidick (1993, 1996), who attribute the late Holocene RSL rise to a down-warping of the crust caused by a neoglacal ice sheet readvance. These authors cite the close agreement between the rise in RSL inferred from the drowned Norse ruins at Sandnes and the general increase in glacier extent in Disko Bugt during the last millennia. This 'Norse' model envisages RSL rising by up to 6 m during the last 0.9 ka cal. yr BP (Weidick, 1993).

Hypothesis formulation

This paper uses isolation basin data from Arveprinsen Ejland, together with other previously published RSL information from Disko Bugt (e.g. Rasch, 1997), to test two hypotheses regarding patterns of post-glacial crustal motions in the study area. Firstly, we test the hypothesis that the post-glacial isobases in Disko Bugt lie parallel to the present ice-sheet margin. This assumption is implicit in two recent analyses of RSL change in Disko Bugt completed by Weidick (1993, 1996). For example, his hypothetical shoreline-relation diagram (Fig. 3) plots shorelines against distance from the inland ice margin and longitude. More recently, Weidick (1996, his fig. 21.2) also proposed three RSL curves for Disko Bugt based on data compiled from north-south transects along the 52°, 53° and 54° W lines of longitude. Both analyses assume that the isobases are parallel to the ice sheet margin and that no significant north-south variations in isostatic rebound exist within the study area.

The second hypothesis examined is that late Holocene RSL rise in Disko Bugt reflects neoglacal ice sheet readvance and renewed crustal loading. Theoretically, this should be manifest by enhanced RSL rise in sites closest to the ice sheet and reduced rates in the more distant outer coastal areas. Moreover, the onset of this rise in RSL should occur earlier in east Disko Bugt compared with the west. Importantly, both these trends are the reverse of those expected to arise from regional forebulge collapse.

Isolation basin litho- and biostratigraphy

Isolation basins are natural rock depressions that may, at various times in their history, be connected to, or isolated from, the sea by changes in RSL. It is common to identify three stratigraphical units within basins that have been isolated from the sea by RSL fall: a basal clastic unit of marine origin, a brackish transitional unit formed as isolation occurs, and an upper freshwater gyttja deposited after isolation (e.g. Foged, 1977; Krzywinski and Stabell, 1984; Retelle *et al.*, 1989; Corner and Haugane, 1993). In small basins where water mixing is high, or where rates of land emergence are rapid, the transitional unit may be thin and the isolation process abrupt. In other deeper basins, massive black sulphidic or finely laminated sediments are common in this transitional unit. These laminations are often paper-thin (e.g. Foged, 1977) and probably reflect reduced bioturbation due to anoxic bottom conditions (Snyder *et al.*, 1997).

Accompanying these stratigraphical changes are up to four biostratigraphical isolation contacts, as defined using diatom analyses (Kjemperud, 1986), and three of these are relevant to this study. The diatomological (or phytological) isolation contact is defined as 'a horizon that was at the sediment-water interface at the time when the water in the photic zone of the sedimentation basin became fresh' (Kjemperud, 1986, p. 64). As RSL falls further, so marine incursions into the basin cease entirely (the hydrological isolation contact). Lastly, the sedimentological isolation contact records the change from a predominantly allochthonous minerogenic sediment of marine origin to an autochthonous freshwater organic deposit.

Also important to isolation basin studies is the prevailing rate of RSL change and the tidal range. Basin isolation will clearly occur fastest in areas where the rate of RSL fall is high and where the tidal range is low. Where the reverse situation exists, however, the isolation process may take place over several hundreds or thousands of years and be recorded by several decimetres to metres of sediment. This can introduce difficulties in identifying the various isolation contacts defined above. One of the advantages of working in Disko Bugt is the low tidal range (ca. 2 m) and, during the early and mid-Holocene at least, the rapid rate of RSL fall.

Techniques

The stratigraphy of each lake was established using a gouge and Russian corer from a small boat, with sediments described using the Troels-Smith (1955) scheme of stratigraphical notation. Sill elevations were determined using a Sokisha level, with all altitudes expressed relative to the mean high water mark (established by daily observation

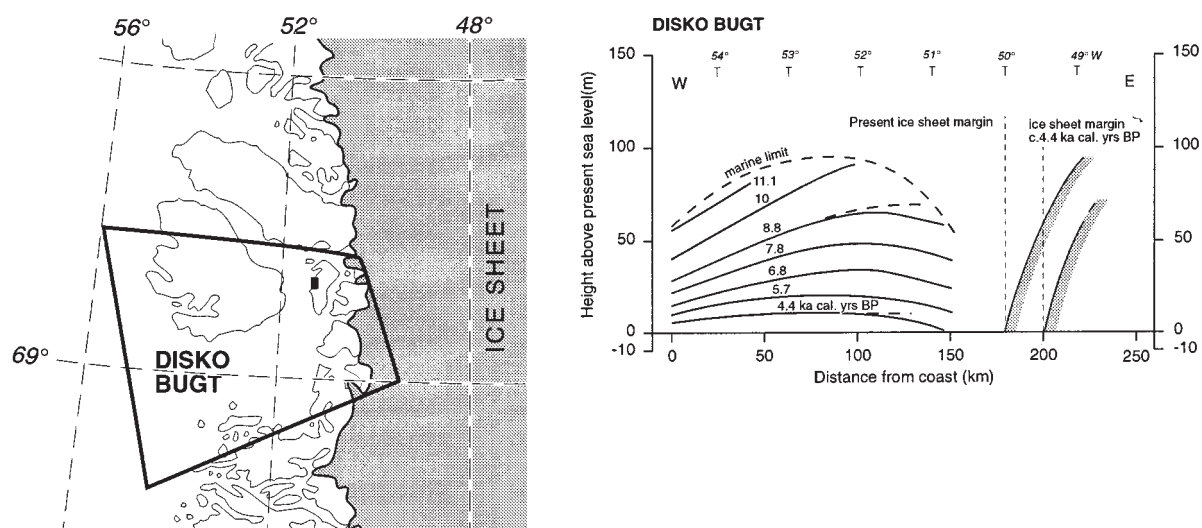


Figure 3 Hypothetical shoreline relation diagram for Disko Bugt (from Weidick, 1993). Dashed shorelines indicate the suggested trend of the same shoreline before the neoglacial ice sheet readvance. The position of the Greenland Ice Sheet at ca. 4.4 ka cal. yr BP, prior to this readvance.

during the study period), subsequently reduced to mean sea-level using tide tables (Farvandsvæsenet, 1995). At each basin one or more levelling transects was completed across the sill, with multiple observations used to determine the maximum and minimum elevations. In some instances the basin sill was clearly defined, comprising unshattered bedrock, but in others it consisted of a wide area (up to 200 m) of shattered bedrock, which was, in places, partially covered by sediment or vegetation. In the latter instances a gouge corer was used to establish the minimum sill elevation. The total closing error in the levelling survey was 0.06 m.

Samples for diatom analysis were prepared using standard techniques (Palmer and Abbot, 1986) and diatom identifications were made with reference to Van der Werff and Huls (1958–1974), Hustedt (1957), Patrick and Reimer (1966, 1975), Hendey (1964) and Foged (1972, 1973, 1977). The diatoms are grouped according to a simplified halobian classification scheme of Hustedt (1957), with frequencies expressed as a percentage of total diatom valves (%TDV). A minimum of 300 diatom valves was counted at most levels. For visual interpretation only those taxa attaining values > 2% TDV are shown on the diatom diagrams. Full counts are available from the lead author.

A combination of AMS and conventional radiocarbon dates were used in this study (Table 1), with new dates reported here (Table 1) based on HCL pre-treated bulk sediment. Not all cores were dated owing to the rapid rate of RSL fall and also the possibility of sediment slumping observed in some of the basin stratigraphies (e.g. basin V1,

see below). Limited amounts of basal marine sediment were retrieved during coring and the only shells contained within these were not *in situ* and therefore were unsuitable for dating. Where possible we collected core material using a 5 cm diameter piston corer, but in some instances water depth and sediment composition necessitated the use of a smaller diameter (4 cm) Russian sampler. In the latter circumstances, AMS dating of bulk (usually 1 cm thick) sediment slices was the best dating option available. The radiocarbon dates presented here are given in 1000 calibrated years before present (ka cal. yr BP) and 1000 radiocarbon years BP (ka ^{14}C yr BP). All marine dates are taken from Rasch (1997) and have been normalised to 0.0‰ PDB before calibration. The calibration program used is Calib 3.03 (Stuiver and Reimer 1993) and all calibrated dates are cited with a 2σ age range.

Results

The litho- and biostratigraphy of each basin studied is shown in Figs 4(a–c) and 5(a–h). Each basin sequence is now discussed.

Basin V0

Basin V0 is the highest of the basins studied and formerly was connected to the sea to the southwest (Fig. 4a). The

Table 1 List of radiocarbon ages from Vaskebugt, Arveprinsen Ejland, Disko Bugt (West Greenland)

Basin	Laboratory code	^{14}C age $\pm 1\sigma$	$^{13}\text{C}/^{12}\text{C}$ ratio	Cal. yr BP $\pm 2\sigma$	Maximum sill altitude (m above MSL)	Minimum sill altitude (m above MSL)	Mid-sill altitude (m above MSL)	Isolation contact	Reference water level	Indicative meaning	MSL	Type of date
V0	Beta-107879	8820 $\pm$ 100	–24.3	9987–9525	69.71	69.30	69.5	D, H, S	MHWST to HAT	1.05 $\pm$ 0.46	68.45 $\pm$ 0.46	Conventional
V2	Beta-112544	7780 $\pm$ 120	–31.0	8956–8329	47.39	46.64	47.01	H, S	HAT	1.10 $\pm$ 0.38	45.91 $\pm$ 0.38	Conventional
V5	Beta-112543	6950 $\pm$ 150	–24.7	8053–7476	25.63	25.19	25.41	D, H, S	MHWST to HAT	1.05 $\pm$ 0.78	24.36 $\pm$ 0.78	Conventional
V7	Beta-110748	6290 $\pm$ 40	–21.3	7235–7038	18.31	18.01	18.16	D, H, S	MHWST to HAT	1.05 $\pm$ 0.25	17.11 $\pm$ 0.25	AMS
V8	Beta-110747	4510 $\pm$ 50	–24.4	5310–4980	14.75	13.29	14.02	D, H, S	MHWST to HAT	1.05 $\pm$ 0.93	12.97 $\pm$ 0.93	AMS
V9	Beta-110457	3750 $\pm$ 130	–26.7	4503–3721	–1.40	–2.10	–1.75	D	MHWST	1.00 $\pm$ 0.37	–2.75 $\pm$ 0.37	AMS
V9	Beta-110456	1360 $\pm$ 90	–21.4	1407–1066	–1.40	–2.10	–1.75	D, H, S	MHWST to HAT	1.05 $\pm$ 0.37	–2.80 $\pm$ 0.37	AMS

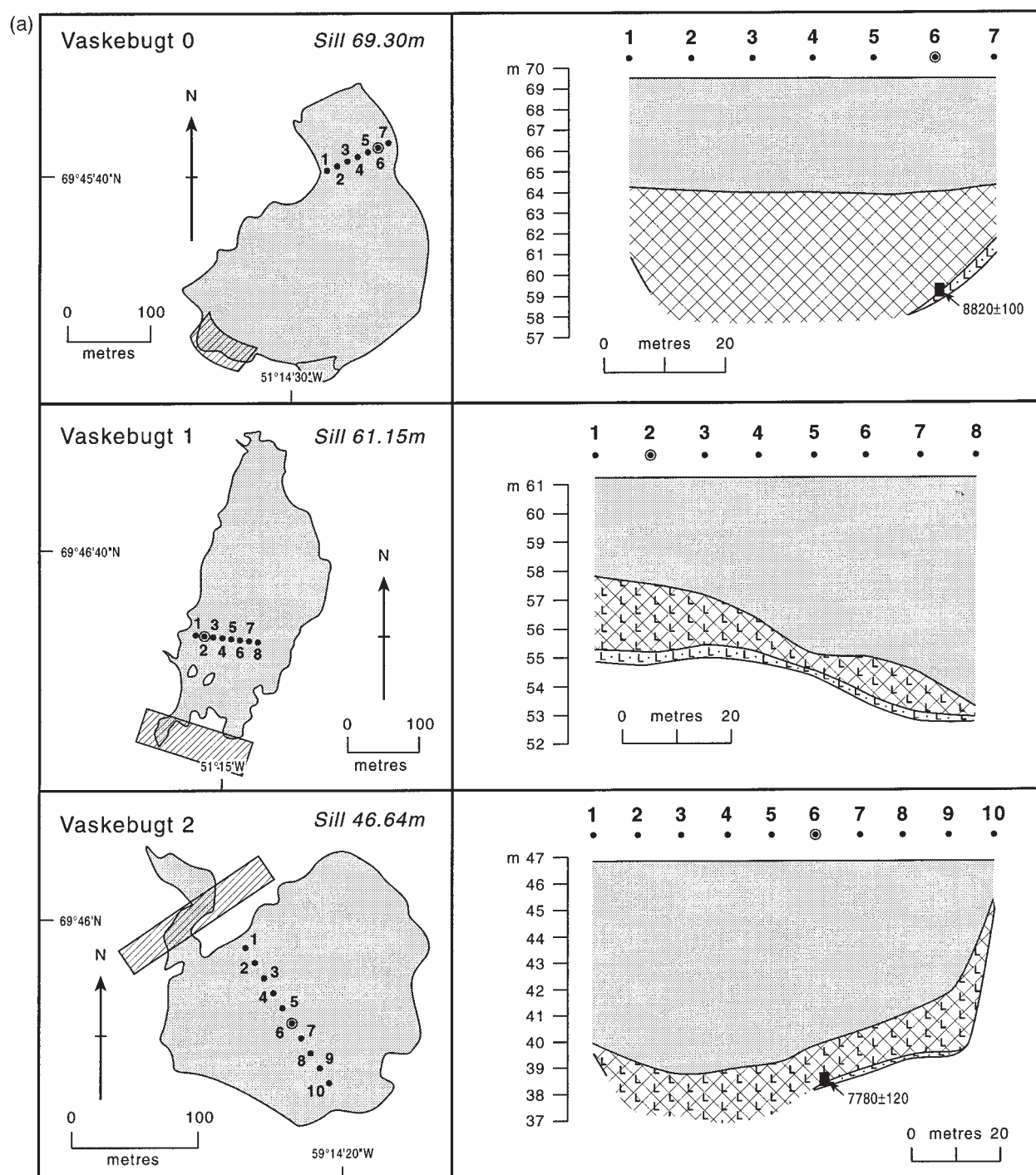


Figure 4 The Vaskebugt isolation basins, showing coring locations of each basin, the position of the levelling transects (hashed area), simplified stratigraphical plots and radiocarbon dates.

basin sill comprises shattered bedrock, which varies in altitude between 69.71 m and 69.30 m. Water depths are shallow in the southwest (< 2 m) but increase to the northeast beneath a steep rock wall, which rises ca. 30 m above the lake level. A transect of seven cores was completed across the embayment in the northeast of the basin (Fig. 4). Maximum coring depth in the basin was 11 m, and in five of the seven cores the gyttja was not penetrated. In the base of cores 6 and 7 a grey clay silt was recorded, separated from the overlying gyttja by a laminated transitional unit. Replicate piston cores were collected of the isolation stratigraphy from core 6.

The lower part of the sample core, between 1048 cm and 1026 cm contains abundant (ca. 80% TDV) polyhalobous

diatoms (notably *Cocconeis costata* and *Cocconeis scutellum*) (Fig. 5a). The diatomological isolation contact is recorded at 1026 cm and above this the clay silt grades into a finely laminated organic silt (1023 cm to 1017.5 cm), similar to the isolation sediments described by Snyder *et al.* (1997). A continued freshening trend occurs above 1019 cm, the sedimentological contact, with a change in lithostratigraphy to a red/brown laminated gyttja. *Fragilaria* spp. is characteristically dominant in recently isolated basins (Stabell, 1985). A sample of the upper part of the laminated organic silt and the overlying freshwater gyttja (1018.5 cm to 1014.5 cm) was dated to 8820 ± 100 yr BP (9987–9525 cal. yr BP, Beta-107879).

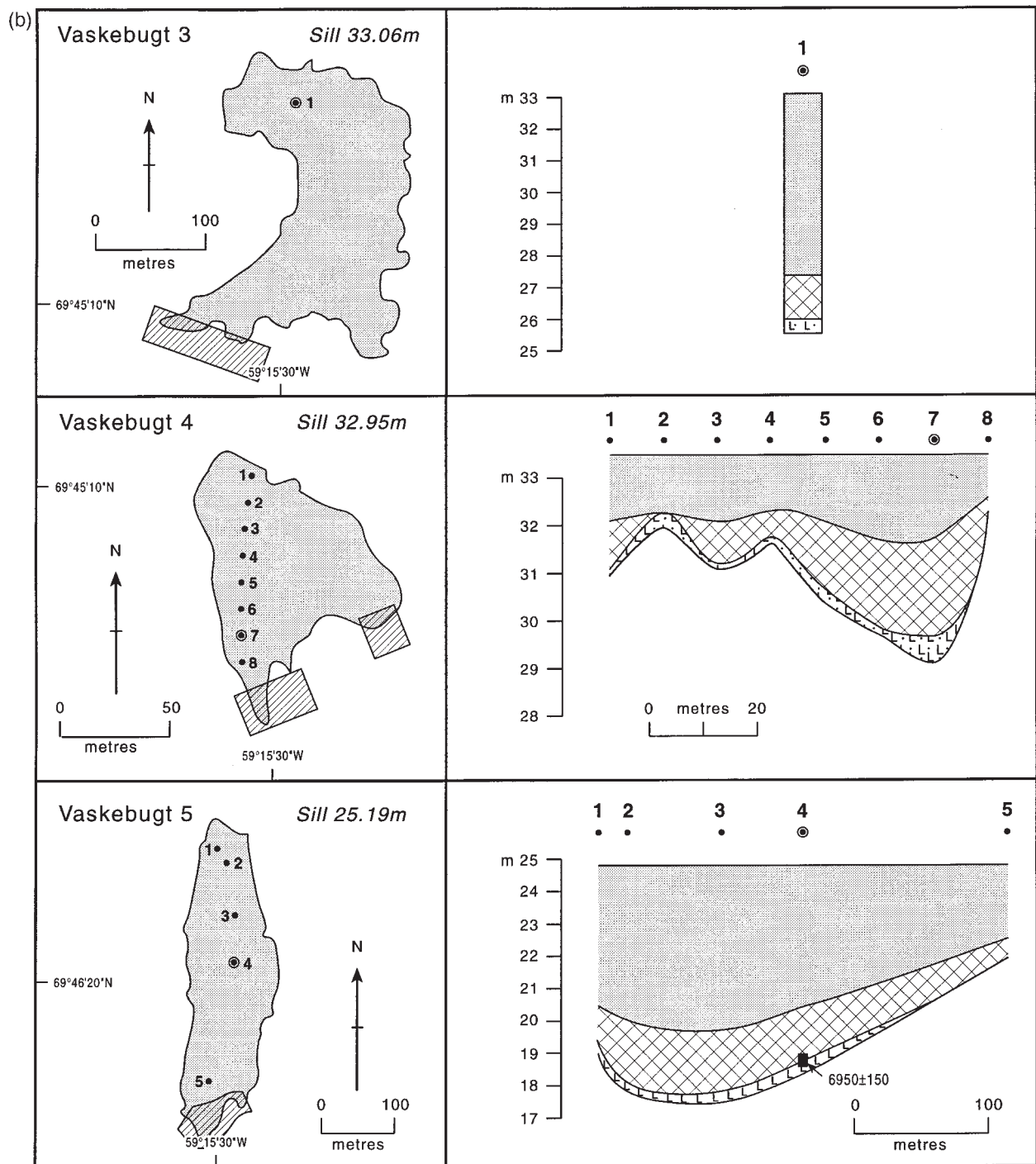


Figure 4 Continued.

Basin V1

Basin V1 opens in a southerly direction across Vaskebugt and its sill comprises shattered bedrock, which varies in elevation between 61.93 m and 61.15 m. A transect of eight cores was completed across the shallower southern part of the basin (Fig. 4a). The basic stratigraphy of the basin is similar to that of V0 except that close to the transition between the lower silts and the overlying gyttja is a prominent horizon of coarse sand and angular gravel, typically 0.5 cm to 2 cm thick. In some cases this deposit lies immediately above the lowermost deposit of gyttja (e.g. core 2), but in others it occurs within the base of the gyttja itself (e.g. cores 1 and 3). Because of the non-cohesive nature of the sediments in this basin, sediment retrieval was not always

good and this complicated efforts to determine the exact stratigraphical position of the coarse unit within the isolation sediment sequence. A piston core of the isolation sediment sequence was collected for analysis from the position of core 2.

The diatom data reveal a marine origin for the lower silt clays, which contain high frequencies (ca. 60%) of polyhalobous taxa including *Cocconeis costata* and *Diploneis crabro* (Fig. 5b). Diatom frequencies are very low in the two levels between 513 cm and 508 cm, where it was not possible to achieve a count in excess of 300 valves. Between 508 cm and 504 cm, frequencies of oligohalobous taxa rise to dominate the assemblage. Above 503.5 cm the minerogenic content of the core decreases and the diatom assemblage becomes characterised by the oligohalobous taxon

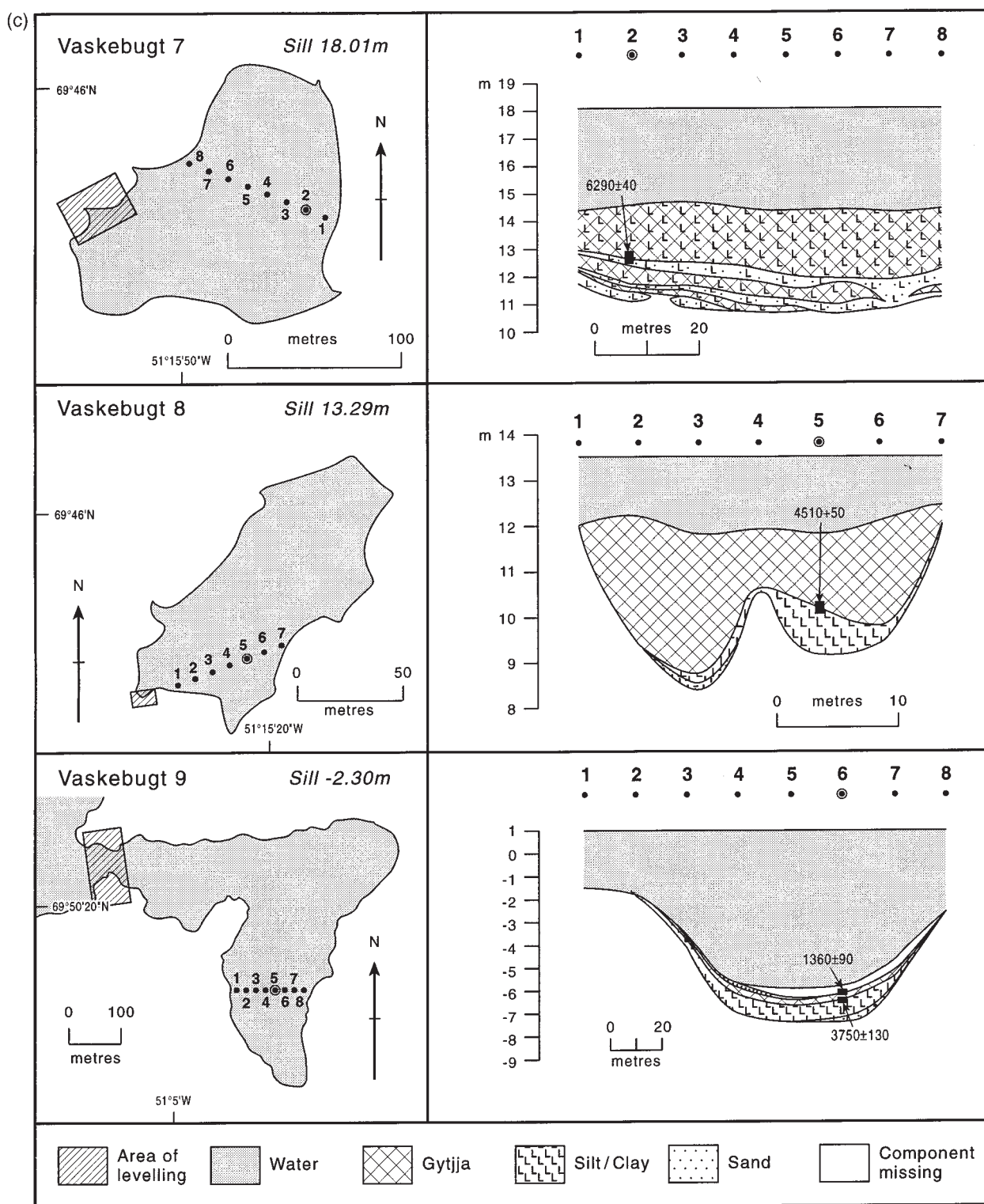


Figure 4 Continued.

Fragilaria construens. The low diatom counts beneath the sedimentological isolation contact (between 513 cm and 508 cm), coupled with the high frequencies of the robust taxa *Diploneis crabro* between these levels, suggest dissolution of the diatom assemblage in this part of the core. This, together with the coarse sand (506.5 cm to 505.5 cm), suggests that a hiatus in sedimentation occurred during the isolation process. A similar thin sand layer is reported by Fredskild (1995) in a high Arctic Greenland lake, where bio- and chronostratigraphical analyses reveal a pronounced

hiatus caused by sediment slumping. The isolation contact in V1 has not been dated.

Basin V2

This basin occupies a large semi-circular depression open to the northeast (Fig. 4a). The lake sill is a 200 m wide flat area of partly vegetated shattered bedrock. Two levelling

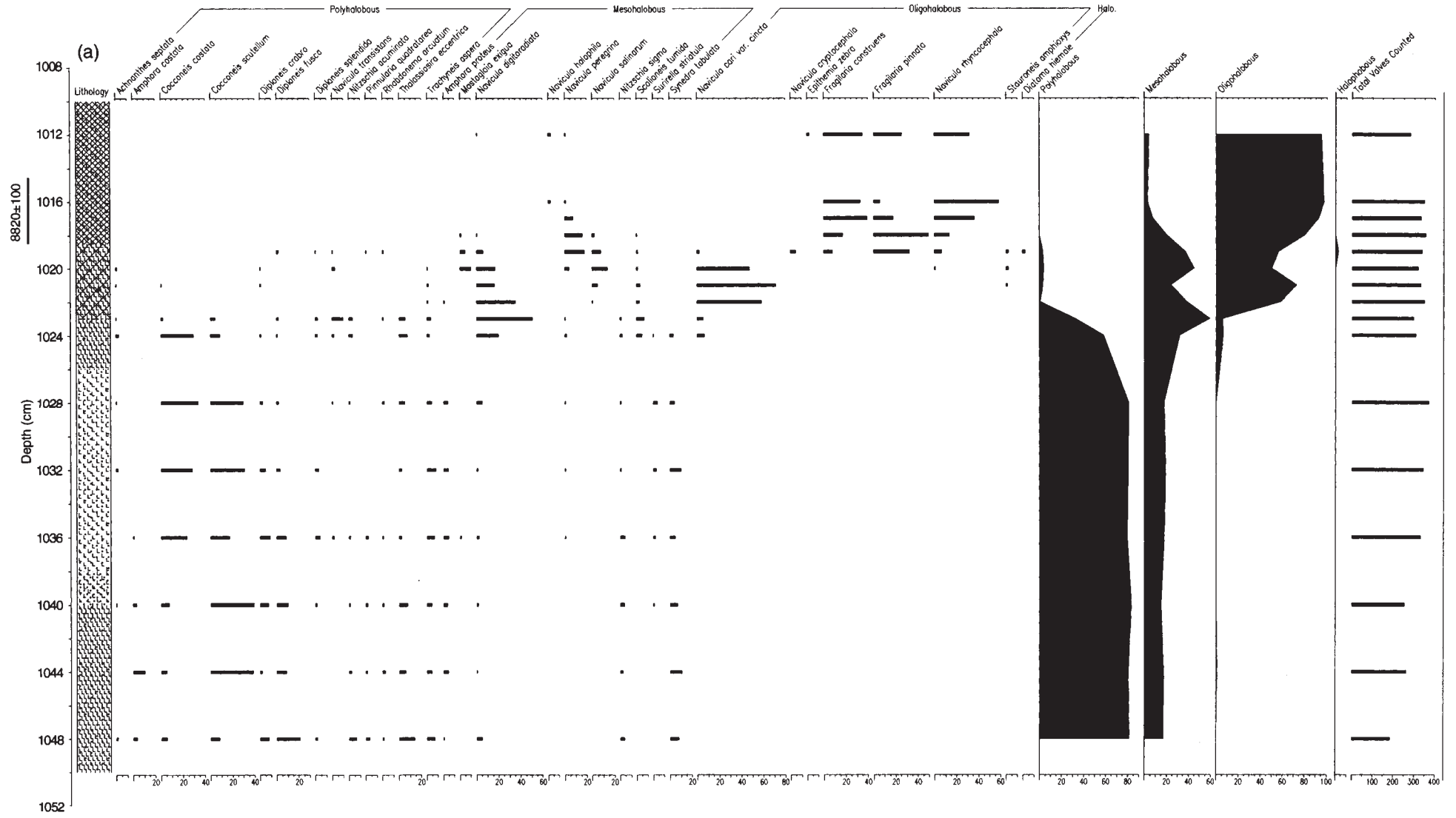


Figure 5 (a) Diatom diagram from V0. Diatoms are expressed as a %TDV, with only taxa > 2%TDV shown. Lithological symbols are after Troels-Smith (1955). Dates are in radiocarbon years BP.

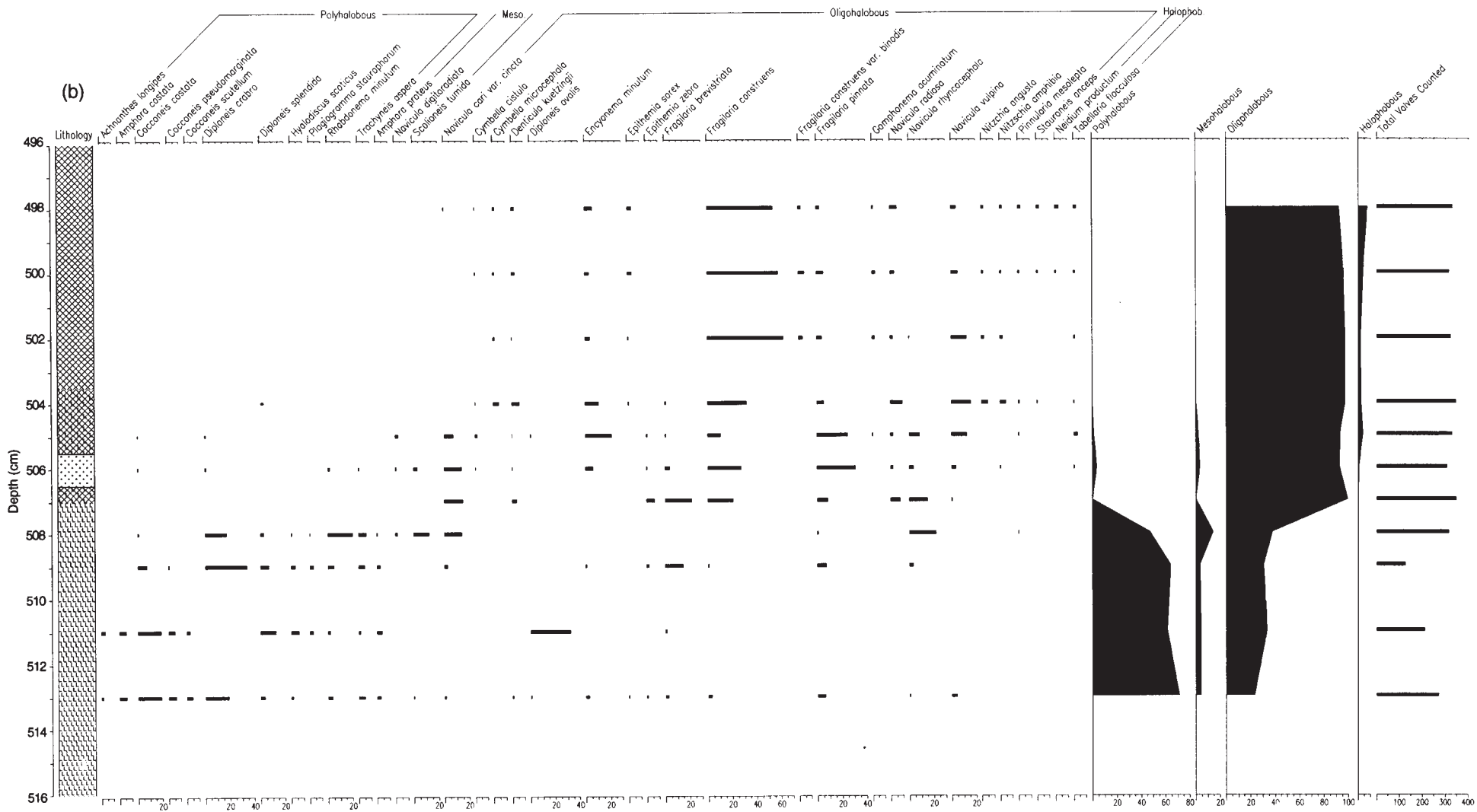


Figure 5 (b) Diatom diagram from V1.

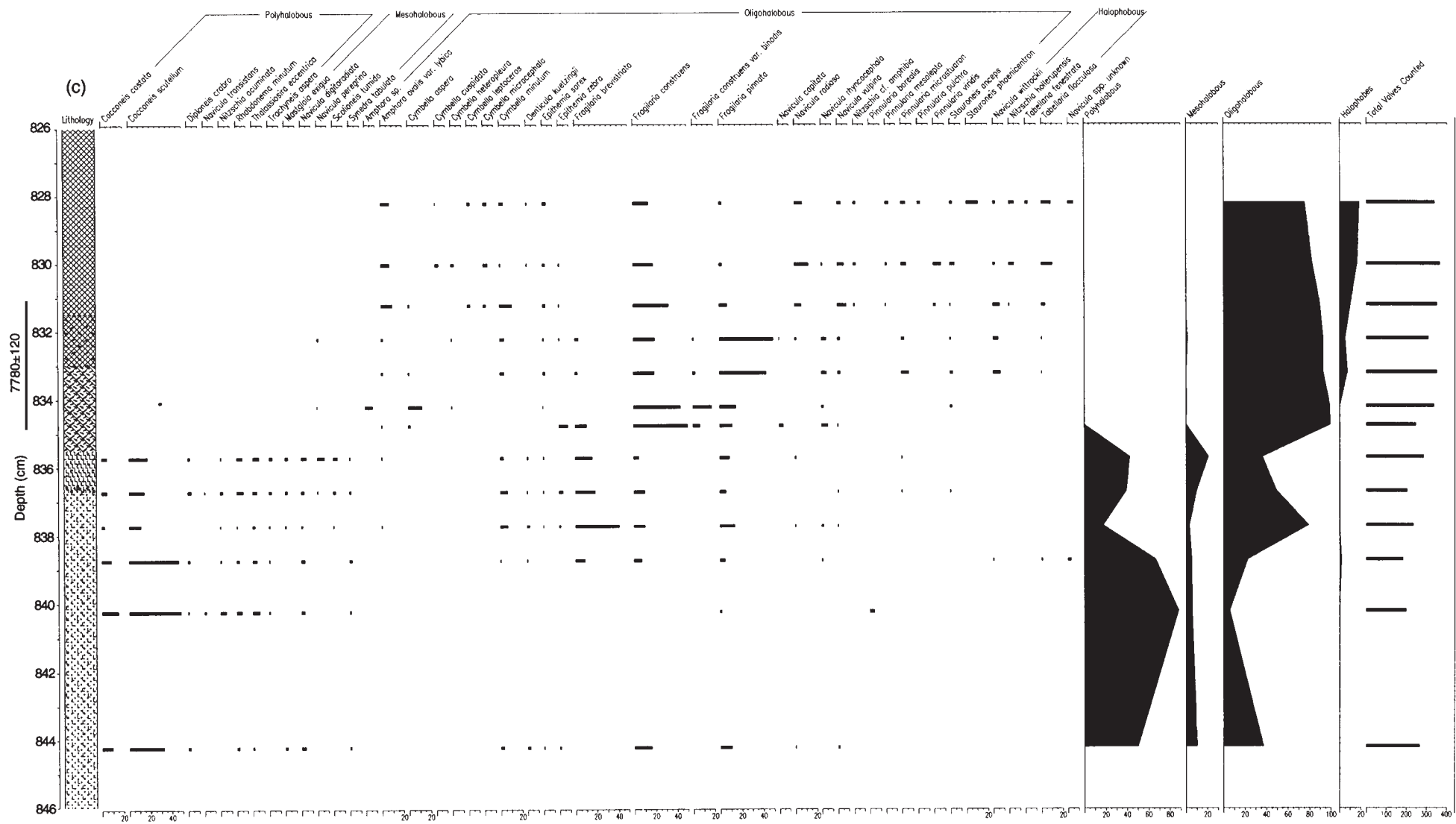


Figure 5 (c) Diatom diagram from V2.

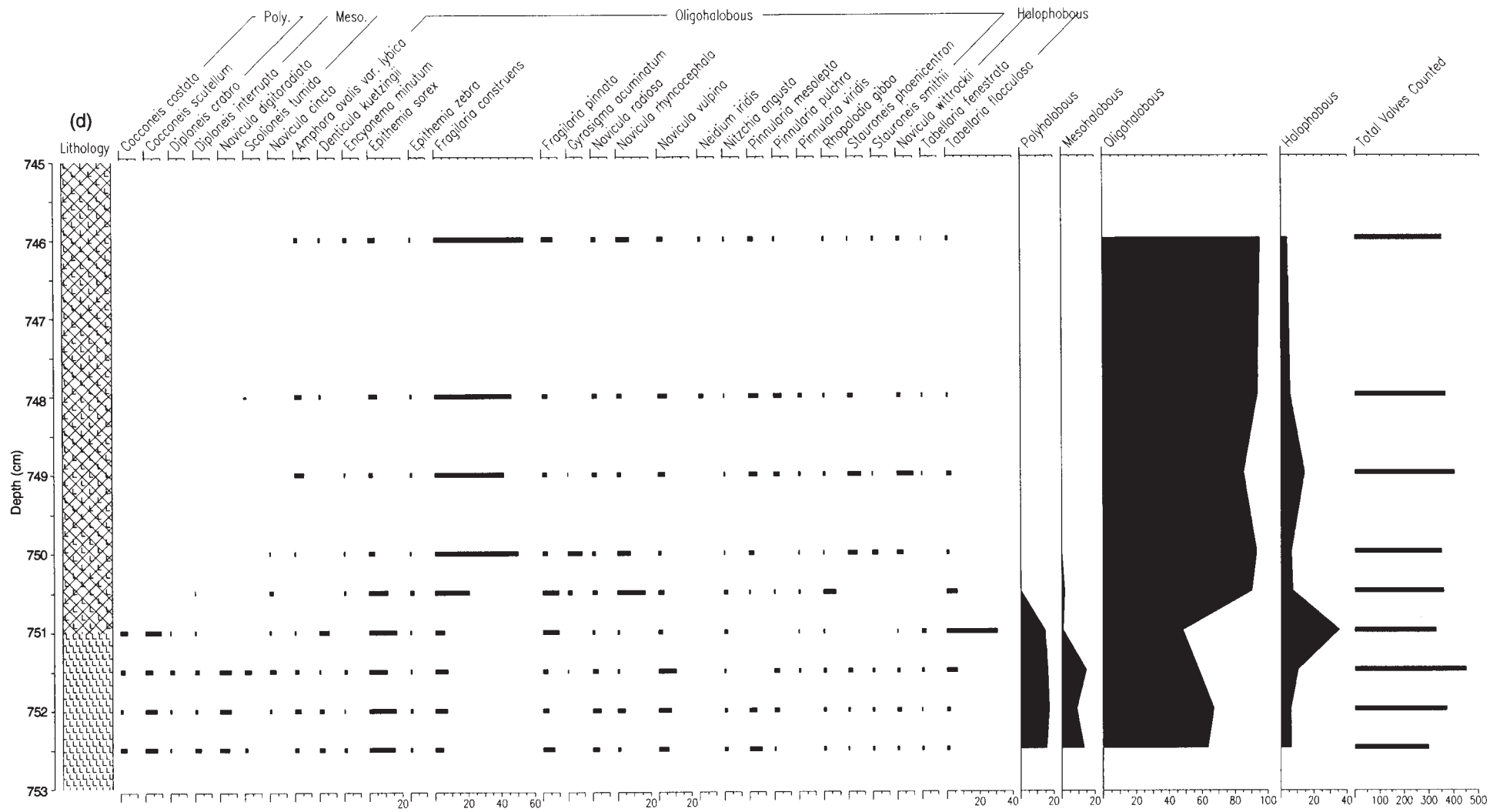


Figure 5 (d) Diatom diagram from V3.

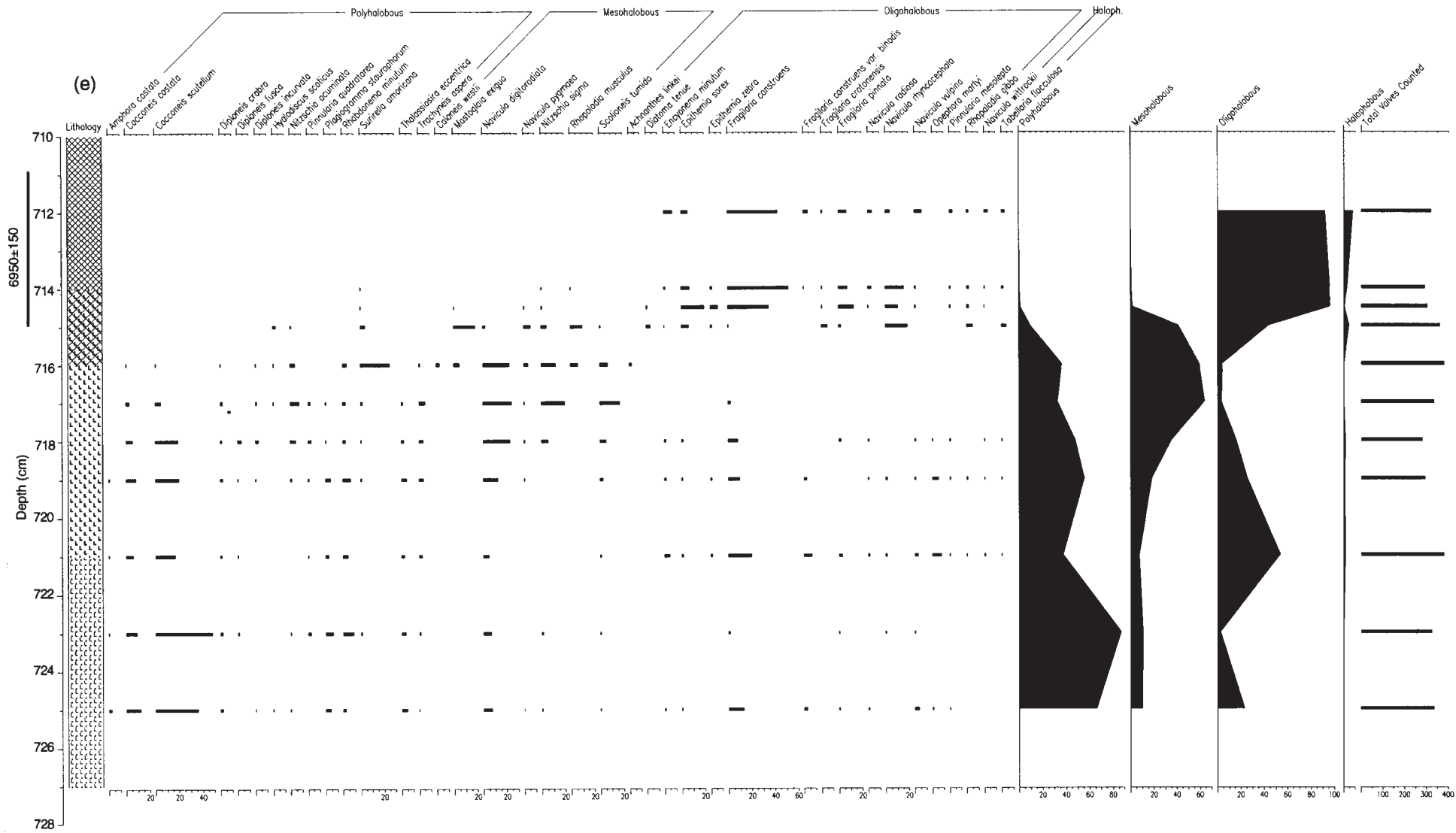
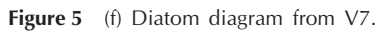


Figure 5 (e) Diatom diagram from V5.



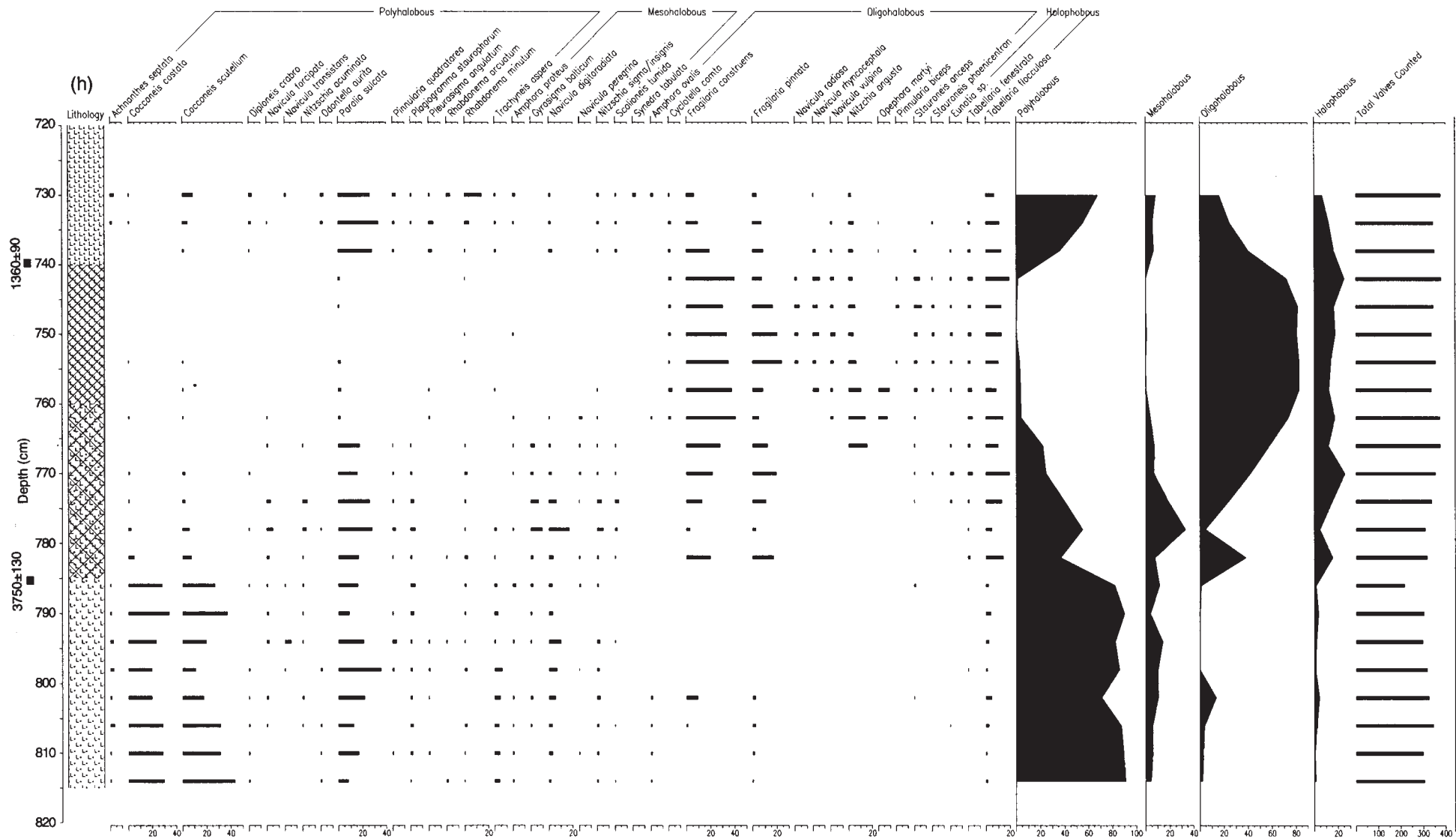


Figure 5 (h) Diatom diagram from V9.

transects across the sill of the basin indicate a height range of between 47.39 m and 46.64 m. The stratigraphy of the basin was determined with a transect of 10 boreholes. Water depths in the basin were between 5 m and 7 m. As in V0 and V1, we were unable to penetrate the gyttja in all cores, although in cores 6 to 9 we recorded a transitional laminated organic silt between the lake gyttja and an impenetrable grey silt-sand. A piston core was collected from the position of core 6 for laboratory analysis.

The base of the sample core in V2 (846 cm to 836.5 cm) is an olive-grey medium-grained sand with some silt and rare detrital herbaceous plant matter. This deposit fines upwards into a grey silt with some clay and then, between 835.5 cm and 833 cm, a laminated transitional unit which passes into the main deposit of gyttja. Diatoms within the lowermost sediments (845.5 cm to 835.5 cm) record a replacement of marine by more brackish conditions (Fig. 5c). A pronounced decline in poly- and mesohalobous taxa occurs at the base of the laminated transitional unit, as frequencies of oligohalobous taxa (*Fragilaria construens* and then *Fragilaria pinnata*) rise to dominate the assemblage. This change marks the diatomological and sedimentological contacts. A sample for radiocarbon dating taken from 835 cm to 831 cm yielded an age of 7780 ± 120 yr BP (8956 to 8329 cal. yr BP, Beta-112544).

Basin V3

This crescent-shaped basin lies some 28 m beneath V1 and receives waters draining from the V0 and V1 catchments. The basin lies on a platform 10 m above the valley floor occupied by V4 and V5. The final connection to the sea was via the southwest arm of the basin, which has a height range of 34.02 m to 33.06 m. A combination of deep water and stiff gyttja meant only one core was collected from the northeastern arm of the basin (Fig. 4b). The stratigraphy of the sample core consists of a small amount of grey silt in the base (753 cm to 751 cm), above which is a laminated brown organic silt (751 cm to 748.5 cm) and then a silty lake gyttja.

Diatoms from the basal grey silt include polyhalobous and mesohalobous taxa in low frequencies (ca. 20% and 15% TDV respectively) (Fig. 5d). Comparatively high (50% to 60% TDV) frequencies of oligohalobous taxa occur in these lower sediments, notably *Epithemia sorex* (ca. 15% TDV). At the base of the transitional laminated unit frequencies of polyhalobous and mesohalobous taxa fall to trace levels and the assemblage becomes dominated by oligohalobous taxa with high frequencies of *Fragilaria construens* (40% TDV). No samples have been dated from this core, but the diatom evidence confirms RSL fall during this period. Comparison with the age determinations from higher and lower basins suggests that isolation of this basin occurred at ca. 8.1 ka cal. yr BP (7.2 ka ^{14}C yr BP).

Basin V4

Basin V4 lies in a small corrie surrounded by steep headwalls that open to the southeast. There are two sills to the basin separated by a bedrock knoll. These sills have a height range of 34.40 m to 33.95 m. A transect of cores across the lake (Fig. 4b) revealed the deepest sequence of sediments in the southwest. Organic sediments (*turfa* and *detritus*

herbaceae) found in the base of cores in the shallow portion of the lake were probably buried by slopewash. Because the elevation of the basin is similar to V3, no detailed laboratory analyses of the sediments were completed.

Basin V5

This basin lies below V3 and V4 and has a 200-m-wide sill made of partly shattered bedrock (Fig. 4b). The sill varies in elevation between 25.63 m and 25.19 m. Five boreholes from the basin reveal an impenetrable clay silt (725 cm to 716 cm) overlain by a laminated transitional unit (716 cm to 714 cm) and then lake gyttja. A sample core was retrieved from the position of core 4 for laboratory analysis.

Diatoms from the basal grey minerogenic sediments are dominated by polyhalobous taxa, although lesser frequencies of mesohalobous and oligohalobous taxa are also present (Fig. 5e). Between 718 cm and 715 cm, polyhalobous taxa decline as those of *Navicula digitoradiata*, *Nitzschia sigma* and *Scolioneis tumida* rise. The laminated transitional unit is marked by a further decline in frequencies of polyhalobous and mesohalobous taxa to trace levels, and a rise in values of oligohalobous taxa. This diatomological and sedimentological contact is dated between 715 cm and 711 cm to 6950 ± 150 yr BP (8053 to 7476 cal. yr BP, Beta-112543).

Basin V6

This basin is a seasonally flooded depression with *turfa*-rich sediments (Fig. 1e). Debris fans flowing from the northeast slopes above the basin appear to have promoted water-logging of this shallow depression. No further analysis of this basin was completed.

Basin V7

Basin V7 is a small basin with debris fans defining the northern limit of the basin and exposed bedrock elsewhere. A stream draining from the combined V3 to V5 catchments enters the north side of the basin and has created a small subaqueous debris fan. The basin was connected to the sea from the southwest, and the sill, over which a small outlet stream drains, comprises intact bedrock with a height range of 18.31 m to 18.01 m. A transect of eight cores was completed across the basin (Fig. 4c) which revealed a complex sequence of deposits with up to three laterally extensive minerogenic units intercalated with more organic-rich sediments. A set of overlapping Russian cores was collected from the position of core 2 for laboratory analysis.

Diatom analyses through the sample core reveal only one isolation contact at 549 cm (Fig. 5f). Polyhalobous taxa dominate the lower part of the sediment core (from 675 cm to 612 cm) but a gradual decline in these taxa occurs above 604 cm as mesohalobous and oligohalobous taxa increase in importance. A further change in diatom stratigraphy takes place above 570 cm as frequencies of the halophobous taxon *Tabellaria flocculosa* rise to 5% to 10% TDV and those of the mesohalobous taxon *Navicula digitoradiata* increase to ca. 20% TDV. Final isolation of the basin, marked by an abrupt decline in frequencies of poly- and mesohalobous taxa, occurs at 549 cm. The AMS dating of a bulk sample

of gyttja from 549 cm to 548 cm yielded an age of 6290 ± 40 yr BP (7235–7038 cal. yr BP, Beta-110748).

Basin V8

This basin, the lowest in the catchment, has a sill altitude of between 14.75 m and 13.29 m. In the deeper parts the lake gyttja is underlain by grey silts and clays (Fig. 4c) and a Russian sample was collected from the position of core 5 for laboratory analysis. In the base of core 3 an impenetrable deposit of dark red/brown, humified *turfa* with some detrital herbaceous matter is overlain by grey silts and clays. In several of the cores the transition from these silts and clays to the overlying gyttja was abrupt (*limes superior* 4) or was marked by a thin (2 mm) bed of sand. These observations suggest that some sediment reworking may have occurred in the basin, with slumping of silts and clays from neighbouring slopes prior to freshwater organic sedimentation.

Diatoms in the grey silts of core 5, between 330 cm and 327.5 cm, confirm their marine origin, with high but falling frequencies of polyhalobous taxa, notably *Cocconeis scutellum* (Fig. 5g). Frequencies of oligohalobous taxa rise continuously through these lower sediments and dominate the assemblage above 327.5 cm, where a sudden decline in poly- and mesohalobous taxa occurs. The AMS dating of a 1 cm bulk sample from the sedimentological and diatomological contact at 327.5 cm yielded an age of 4510 ± 50 yr BP (5310 to 4980 cal. yr BP, Beta-110747).

Basin V9

This basin lies ca. 10 km north of Vaskebugt on the eastern margin of Kangerluk (Fig. 1c and d) and was selected to collect data relating to late Holocene RSL change. The tidal inlet into the basin is 20 m wide and comprises bedrock swept clean by strong tidal currents which accompany the falling and rising tide. The submerged sill height (between -1.6 m and -2.3 m) was measured at slack water from a boat using a levelling staff. A transect of eight cores was completed across the southern part of the basin (Fig. 4c) and revealed the presence of five main stratigraphical units: a lower organic silt with some sand lenses, a grey organic silt, a black gyttja with occasional laminations to base, and an organic silt capped by a coarse lag of sand and gravel. A set of overlapping Russian samples was collected from the position of core 5 for laboratory analysis.

The diatom diagram (Fig. 5h) shows that the lowermost part of the sequence accumulated under marine conditions, as indicated by the high frequencies of polyhalobous taxa (ca. 80% TDV). Between 785 cm and 740 cm there is a pronounced freshening of the basin, with a fall in polyhalobous taxa and a rise in oligohalobous diatoms to account for up to 80% TDV. Of the latter the most important are *Fragilaria construens* and *Fragilaria pinnata*, which account for ca. 40% and 15% TDV respectively. Frequencies of halophobous taxa also rise in importance during this period, as *Tabellaria flocculosa* increase to between 10% and 15% TDV. Above 740 cm there is an abrupt return of brackish conditions to the basin, as indicated by the sharp rise in frequencies of *Paralia sulcata* and other polyhalobous taxa. The AMS dating of two 1 cm bulk sediment samples provide bracketing ages for the isolation and ingression of the basin of 3750 ± 130 yr BP (4503–3721 cal. yr BP, Beta-110457)

and 1360 ± 90 yr BP (1407–1066 cal. yr BP, Beta-110456) respectively.

Discussion

Holocene relative sea-level change on Arveprinsen Ejland

The data described above enable the development of a local record of RSL change that spans the full post-glacial period based on isolation basin data alone. Definition of the age and altitude errors associated with each index point is an important first step before testing the two hypotheses stated earlier in this paper.

Shennan *et al.* (1994) suggest that the diatomological and hydrological isolation contacts record reference water levels of mean high water of spring tides (MHWST) and highest astronomical tide (HAT) respectively. At Ilulissat, the closest port to Vaskebugt, these datums are 1 m and 1.1 m above MSL. Secular variations in MSL, as well as possible seasonal variations in the M2 amplitude, mean that these predictions should be viewed as accurate to within ca. ± 0.1 m (Woodworth, P., Proudman Oceanographic Laboratory, May 1998, pers. com.). Where the dated sample spans both the hydrological and diatomological contacts, the reference tide level is MHWST to HAT. For each index point the indicative range is calculated as the root mean square error arising from estimates of the sill elevation (which varies between basins), tidal range (± 0.1 m), and the levelling (± 0.03 m) (Table 1). For the sill elevations the error term is calculated using the range between the possible maximum and minimum sill altitude (Table 1). The age of each index point is shown as the mean plus the two sigma calibrated age range of the radiocarbon date (Fig. 6).

The highest index point, from V0, closely dates the age of the marine limit in Vaskebugt. The local marine limit here is defined around the rock slopes that surround V0 by the lower limit of perched boulders. Twenty-one measure-

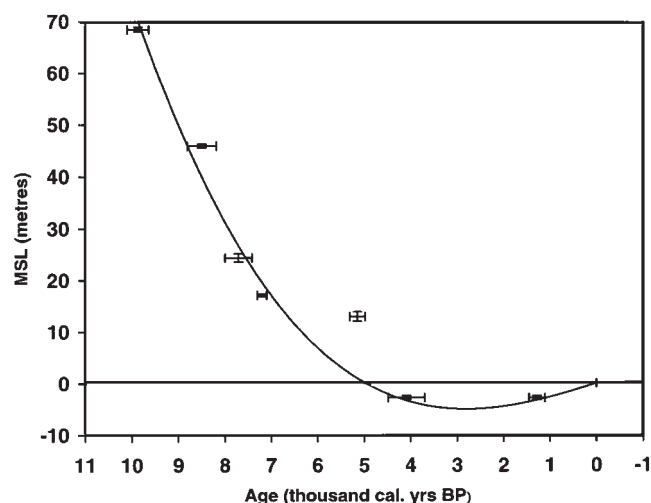


Figure 6 Relative sea-level graph depicting mean sea-level change at Vaskebugt, Arveprinsen Ejland. The y-axis error bars on each index point represent an error term that includes the height range of each sill and errors associated with the levelling and tide gauge data. The x-axis error bar is equal to the 2σ calibrated age range of each index point (Table 1). The smoothed line is a best-fit polynomial, ignoring the V8 index point.

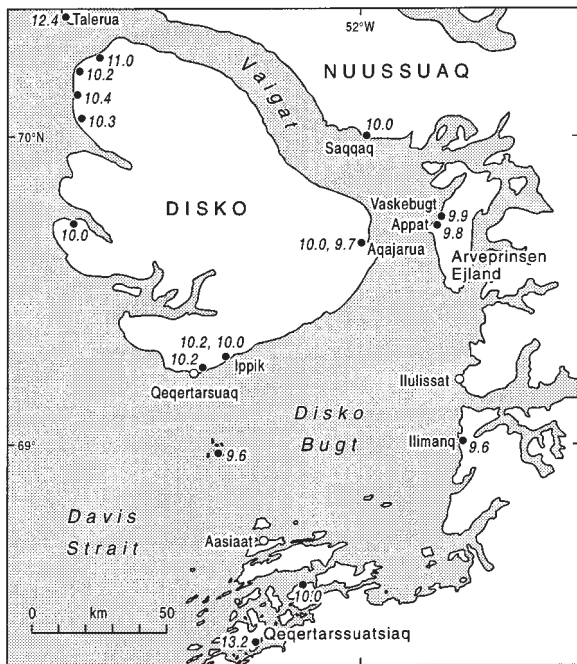


Figure 7 Early Holocene radiocarbon dates (in 1000 calibrated years BP) pertaining to the deglaciation of the Disko Bugt (Table 2). Shell dates have been corrected for a marine reservoir age of 400 yr.

ments on the northwest slopes surrounding the basin establish an altitudinal range for this limit of between 73.08 m and 82.07 m (mean = $77.58 \text{ m} \pm 3.54 \text{ m}$). The protected aspect of the basin, which opened to the sea to southwest, suggests wave run-up was probably not high, although the setting may have amplified the effects of exceptional waves, including those formed by iceberg roll. Allowing for an exceptional wave run-up of $\text{MSL} + 3 \text{ m}$, the sill elevation of V0 (69.71 m to 69.30 m) is within 3 m to 4 m of the lowest estimate of the marine limit (73.08 m).

There is no stratigraphical evidence in V0, or any of the other high-level basins in Vaskebugt, for a rise in RSL to the marine limit. Marine sediments were sampled in only

three cores in V0 but in two of these they were underlain by an impenetrable coarse deposit, probably close to bedrock. No basal freshwater sediments were observed. Our data suggest that this part of Arveprinsen Ejland was ice-free and experiencing significant glacio-isostatic rebound (sufficient to offset the rapidly rising 'eustatic' sea-level at this time) by ca. 10 ka cal. yr BP (9 ka ^{14}C yr BP). As noted above, Funder and Hansen (1996) propose that Disko Bugt was ice-free by 11.1 ka cal. yr BP (10 ka ^{14}C yr BP) at least. These new data, however, add to a growing corpus of evidence (Fig. 7) which suggests that the waters of inner Disko Bugt (i.e. east of Aasiaat–Qeqertarsuaq) were ice-free only after ca. 10 ka cal. yr BP (9 ka ^{14}C yr BP) (Weidick, 1972, 1996; Donner and Junger, 1978; Ingolfsson *et al.*, 1990; Rasch, 1997).

From the marine limit, RSL fell rapidly to intersect present MSL at ca. 5 ka cal. yr BP (4.5 ka ^{14}C yr BP). There is no evidence in the lake basins analysed for any oscillations in RSL during this interval, although the vertical sampling interval between basins is wide. Variations in organic and minerogenic sediment, such as those recorded in V7, record fluctuations in productivity within the lake basin during the isolation process and not oscillations in RSL. Interpolation between the two index points from the submerged basin (V9) suggest MSL continued to fall to a minimum of ca. -5 m at ca. 2.8 ka cal. yr BP (2.5 ka ^{14}C yr BP), before rising by an equivalent amount to the present day. A slow-down in the rate of RSL change is also indicated by the increase in thickness of the V9 isolation sediments compared with those from the higher basins. There is no evidence for more than one oscillation in late Holocene RSL, as observed by others in Disko Bugt (e.g. Rasch *et al.*, 1997), and any such oscillation in Vaskebugt must have been less than ca. 2 m to escape registration in the V9 stratigraphy.

Of the data presented above, only the index point from V8 appears anomalously young for its elevation (Fig. 6). The most obvious source of potential error here is contamination during sediment collection (a Russian sampler and not piston corer was used), although the shallow depth of the lake may have made it more susceptible to root penetration (e.g. Kaland *et al.*, 1984). In addition, the stratigraphy of V8 is

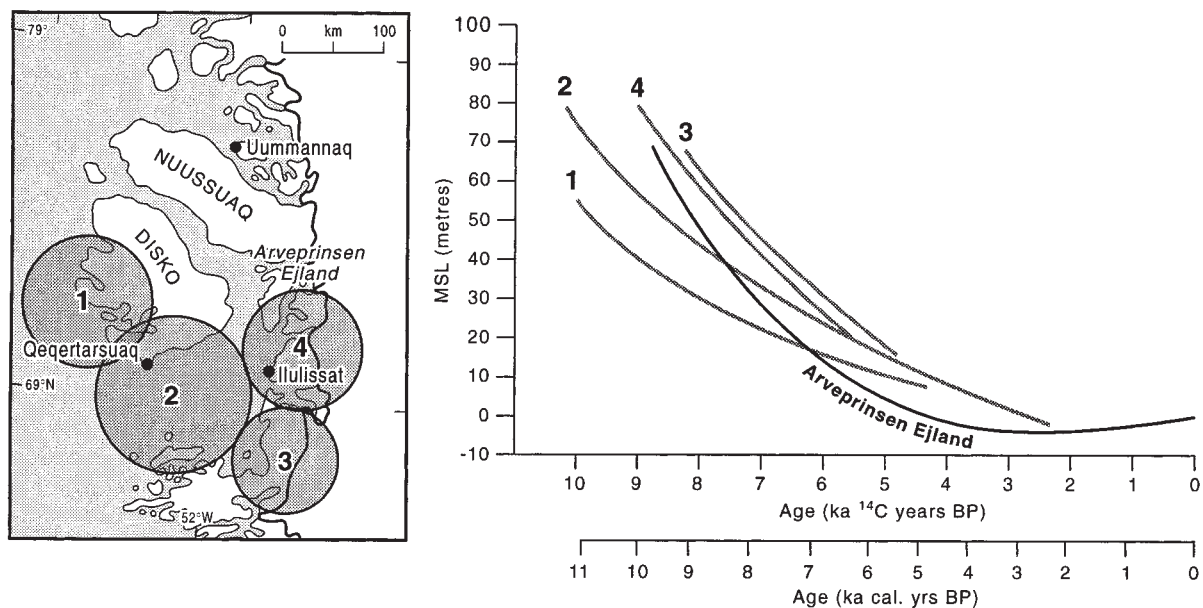


Figure 8 Four relative sea-level curves proposed by Weidick (1996) for Disko Bugt, together with the Arveprinsen Ejland RSL curve generated in this study.

Table 2 List of radiocarbon ages pertaining to the deglaciation of Disko Bugt. Shell dates have been normalised to 0.0‰ PDB prior to calibration and are taken from Rasch (1997). Calibrated dates are given with a 2σ age range

Location	Material	Laboratory code	^{14}C age $\pm 1\sigma$	Cal. max	Cal. mean	Cal. min	Latitude	Longitude	Reference
Qeqertarsuaq	Gytja	K-5133	11320 $\pm$ 140	13575	13230	12936	68°26'	52°57'	Fredksild (1996)
Talerua	Shells	UA-1789	10470 $\pm$ 130	12684	12390	11921	70°23'	54°57'	Bennike <i>et al.</i> (1994)
Niaqussat	Shells	I-16393	9920 $\pm$ 150	12056	11010	10670	70°16'	54°37'	Bennike <i>et al.</i> (1994)
Hammer Dal	Shells	LU-3037	9360 $\pm$ 140	10906	10360	10027	70°10'	54°50'	Ingolfsson <i>et al.</i> (1990)
Inussuup Kua	Shells	K-5510	9350 $\pm$ 100	10797	10330	10041	70°03'	54°45'	Bennike <i>et al.</i> (1994)
Qeqertarsuaq	Shells	AAR-5	9240 $\pm$ 250	10965	10228	9656	69°17'	53°28'	Ingolfsson <i>et al.</i> (1990)
Ippik	Shells	K-4567	9220 $\pm$ 130	10537	10220	9958	69°18'	53°15'	Frich and Ingolfsson (1990)
Illorpaat	Shells	I-16390	9200 $\pm$ 150	10778	10205	9908	70°14'	54°47'	Bennike <i>et al.</i> (1994)
Ivisaarqut	Shells	LU-3041	9060 $\pm$ 90	10283	10010	9895	69°44'	54°50'	Ingolfsson <i>et al.</i> (1990)
Ippik	Shells	HEL-2210	9060 $\pm$ 120	10299	10010	9864	69°18'	53°15'	Ingolfsson <i>et al.</i> (1990), Frich and Ingolfsson (1990)
Kannala	Shells	HEL-362	8970 $\pm$ 170	10301	9970	9529	68°36'	53°34'	Tauber (1960a)
Aqajaru	Gytja	K-3667	8950 $\pm$ 125	10275	9960	9644	69°40'	52°01'	Ingolfsson <i>et al.</i> (1990)
Saqqaq	Shells	K-994	8940 $\pm$ 170	10291	9950	9499	70°04'	52°06'	Tabuer (1960b), Tauber (1964)
Vaskebugt V0	Gytja	Beta-107879	8820 $\pm$ 100	9987	9870	9525	69°45'	51°14'	This paper
Appat	Shells	K-3664	8760 $\pm$ 125	9984	9757	9458	69°44'	51°24'	Ingolfsson <i>et al.</i> (1990)
Aqajaru	Shells	K-3660	8700 $\pm$ 120	9962	9650	9444	69°40'	52°00'	Ingolfsson <i>et al.</i> (1990)
Nunarsuaq	Shells	RCD-21	8690 $\pm$ 90	9903	9620	9457	68°59'	53°19'	Bennike <i>et al.</i> (1994)
Iliminaq	Shells	K-2023	8680 $\pm$ 130	9962	9610	9397	69°01'	51°08'	Tauber (1960b)

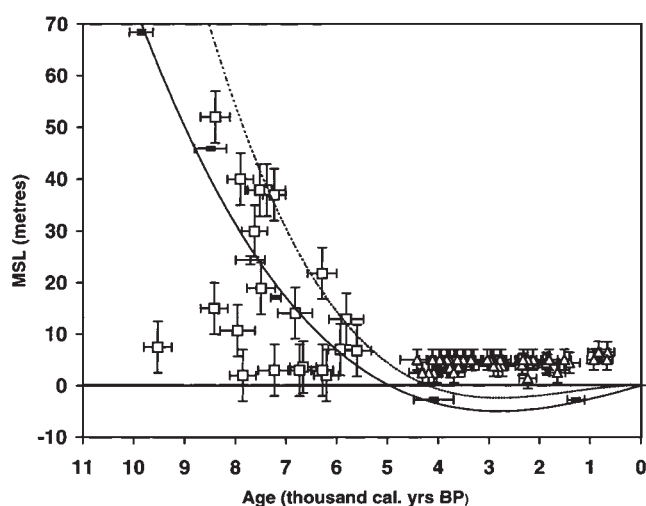


Figure 9 A relative sea-level curve from southeast Disko Bugt (dotted line; Rasch and Jensen, 1997), together with the Arveprinsen Ejland curve generated in this study (solid line). The y-axis error bars are equal to ± 5 m for the shell dates (open square) and ± 2 m for the archaeological and other organic material (open triangles) (Rasch and Jensen, 1997). The error terms for the Arveprinsen Ejland data are defined in Table 1.

variable, suggesting possible sediment reworking. The isolation contact is also abrupt, although this may reflect the small basin size. An alternative interpretation is that the RSL record during this part of the Holocene has a stepped form, although more data are required from basins at ca. 15 m to resolve this issue further.

Hypothesis testing

Isobase orientation in Disko Bugt

As noted above, recent work in Disko Bugt by Weidick (1993, 1996) suggests that post-glacial isobases are parallel to the present ice sheet margin. We now examine this assumption by comparing the RSL data from Arveprinsen Ejland with a series of RSL graphs proposed by Weidick (1996) for Disko Bugt (Fig. 8). In Weidick's analysis, curves 1 and 2 show slower rates of RSL fall during the early and mid-Holocene compared with the curves from areas 3 and

4. These differences indicate important west-east contrasts in post-glacial crustal rebound, with faster uplift in the east compared with the outer coastal areas. In contrast, the close similarity between curves 3 and 4, both from the eastern part of Disko Bugt, suggest minimal north-south variations in crustal rebound within this part of the study area.

When compared with these four records, the Arveprinsen Ejland RSL curve is clearly steeper than those from areas 1 and 2. This confirms the suggestion that rates of RSL fall during the early and mid Holocene are fastest in sites closest to the centre of the Greenland Ice Sheet (e.g. Funder, 1989; Weidick, 1996; Rasch, in press). These differences must also reflect variations in the timing of deglaciation within the study area. Indeed, a plot of the oldest radiocarbon dates from the study area (Fig. 7, Table 2) shows that the few dates > 11.1 ka cal. yr BP (10 ka ^{14}C yr BP) are restricted to the outer coastal area, whereas the oldest dates from within Disko Bugt are all < 10 ka cal. yr BP (9 ka ^{14}C yr BP).

Not all of the radiocarbon ages in Fig. 7 date the marine limit; indeed the age of the marine limit may be only a poor indicator of the timing of local deglaciation, especially if the sea transgressed to its highest point (e.g. Bennike *et al.*, 1994; Funder and Hansen, 1996). Despite these problems, as can be seen from Fig. 7, there is a clustering of early Holocene dates from east Disko Bugt that date to between ca. 9.4 ka and 10 ka cal. yr BP (8.7 ka and 9 ka ^{14}C yr BP). The similarity between these dates and the age of the marine limit on Arveprinsen Ejland (9.9 ka cal. yr BP (8.8 ka ^{14}C yr BP)) implies a reasonably synchronous deglaciation of northeast Disko Bugt at, or shortly after, the end of the Disko Stade (Ingolfsson *et al.*, 1990). This apparent synchronicity raises the possibility of a physical interaction between the Disko Stade glaciers and the retreating Disko Bugt glacier. This issue warrants further research.

Although our results confirm clear west-east differences in RSL across the study area, nevertheless the Arveprinsen Ejland data differ significantly from the curves proposed by Weidick (1996) for areas 3 and 4 (Fig. 8). For example, between 8.8 ka and 7.8 ka cal. yr BP (8 ka and 7 ka ^{14}C yr BP) there is an altitudinal difference between the records of between 10 m and 15 m. To examine this further, the Arveprinsen Ejland data are compared (Fig. 9) with an additional RSL curve from southeast Disko Bugt (similar to area 3 in Fig. 8), which incorporates a combination of recently collected archaeological and sedimentological data (Rasch and Jensen, 1997). A similar altitudinal and chronological offset between the RSL records from the two areas

is observed, which this time is equivalent to ca. 20 m at 7.8 ka to 6.8 ka cal. yr BP (7 ka to 6 ka ^{14}C yr BP).

The causes of these differences in RSL may be attributed to a number of factors. One possibility is that the age – altitude offsets reflect problems inherent in compilations of data from several dispersed sites (see above). Alternatively, the differences may have arisen from contamination of the isolation basin index points by older inwashed carbon (e.g. Retelle *et al.*, 1989) and the reliance on bulk radiocarbon dates (see Björck *et al.*, 1998), although the bedrock geology of Arveprinsen Ejland is dominated by gneisses which should help reduce such problems. Equally, contamination of the shell dates by young carbon seems unlikely given the depositional settings of much of the shell material dated, as does possible uncertainty in the marine reservoir correction factor applied to the shell dates (Rasmussen and Rahbek, 1996). A more compelling explanation for these differences is that they reflect greater isostatic rebound in southeast compared with northeast Disko Bugt. If correct, this interpretation indicates that the isobases in eastern Disko Bugt are, in fact, not parallel to the present ice-sheet margin and that north–south variations in post-glacial RSL change are a significant component of the area's isostatic history.

Three factors may explain the greater isostatic rebound and RSL fall in the southeast part of the bay. Firstly, these differences may reflect a more recent date for deglaciation here compared with areas to the north. If correct, this would imply diachroneity in the retreat of the Greenland Ice Sheet to the eastern shores of Disko Bugt. Testing this hypothesis is difficult, for despite extensive sampling of the raised beaches and spreads of glaciomarine sediments in southeast Disko Bugt (e.g. Donner and Junger, 1975; Rasch and Jensen, 1997) the highest occurrence of dated shells is at only ca. 55 m, some distance below the marine limit, which here locally exceeds 100 m (Rasch, in press).

A second possible explanation is that there was increased ice load in southeast Disko Bugt compared with areas to the north. This scenario would perhaps be compatible with a large ice stream occupying southern Disko Bugt, drawing significantly on the ice discharged from Jakobshavn Isbræ, the largest ice stream in West Greenland. However, when viewed in a wider spatial context, it seems likely that the enhanced uplift in southeast Disko Bugt relates to the northern edge of the dome-shaped area of maximum uplift centred over the ice-free coastal land immediately south of Disko Bugt (e.g. Weidick, 1992; Fig. 2). This doming is presumed to reflect the earlier deglaciation and greater ice loading of the outer coastal zone compared with areas further north. The moraines here, however, are poorly developed and maps of ice margin recession are justifiably cautious (e.g. Funder, 1989). No systematic investigations of RSL change have taken place in this area, although a single date from an isolation contact at 95 m on Qeqertarsuatsiaq, south of Aasiaat (Fig. 7), may provide a minimum date (13.2 ka cal. yr BP (11.3 ka ^{14}C yr BP)) for deglaciation (Fredskild, 1996).

In summary, of the explanations proposed above it seems most likely that the RSL differences observed in Fig. 9 reflect contrasts in the distribution of the ice load and the timing of deglaciation. To discriminate between these alternative explanations it would be useful to date the marine limit at several sites in a west–east transect across Disko Bugt. In addition, future work should seek to date basins immediately above and below the marine limit so as to test the hypothesis that the age of the marine limit in this region is synchronous with deglaciation.

Late Holocene RSL rise

The second hypothesis tested is that late Holocene RSL rise reflects neoglacial ice sheet readvance and renewed crustal loading (Kelly, 1980; Weidick, 1993, 1996). The Arveprinsen Ejland RSL curve intersects present MSL at ca. 5 ka cal. yr BP (4.5 ka ^{14}C yr BP). This is similar in timing to the RSL curves from southeast Disko Bugt and southeast Disko, which intersect present MSL at ca. 5.4 ka and ca. 4.4 ka cal. yr BP (4.7 ka and ca. 4 ka ^{14}C yr BP) respectively (Rasch and Nielsen, 1995; Rasch and Jensen, 1997). Although RSL data are limited from the outer coast of Disko Bugt, nevertheless Rasch *et al.* (1997) argue that on west Disko RSL fell to intersect MSL at ca. 2.7 ka cal. yr BP (2.5 ^{14}C yr BP). This chronological difference across the study area is compatible with the hypothesis that late Holocene RSL rise was caused by early neoglacial ice sheet readvance, and is contrary to what one would expect if the rise in late Holocene RSL was a function of regional forebulge collapse. Rasch *et al.* (1997) note these chronological differences and suggest that they can perhaps be explained by tectonic movements or the local isostatic effects of ice caps on Disko.

Unfortunately, present geophysical rebound models for the region offer little help in resolving these issues further. For example, the ICE-4G global model of post-glacial RSL change proposed by Peltier (1996) defines an area of high present-day RSL rise (between 0 mm and $> 6 \text{ mm yr}^{-1}$) to the west of Disko Bugt in Baffin Bay, caused by the combined collapse of the Greenland and former Laurentian Ice Sheet forebulges. The predicted influence of this forebulge however, does not extend as far east as Disko Bugt, where the ICE-4G model estimates present day RSL fall of between 0 mm and 4 mm yr^{-1} . These predictions are contrary to the empirical evidence for recent RSL rise described above from Disko Bugt.

In summary, therefore, we conclude that at present it is not possible to reject the hypothesis that late Holocene RSL rise reflects neoglacial ice sheet readvance and renewed crustal loading. Further testing of this hypothesis, and others relating to the local effects of ice caps on Disko and tectonic influences, requires the collection of additional RSL data from the outer coastal areas of Disko Bugt to complement the record previously developed from west Disko by Rasch *et al.* (1997).

Conclusions

The Greenland Ice Sheet is a key contributor to past and future climate and sea-level change (Clark and Primus, 1987; Gregory and Oerlemans, 1998), and the intimate link that exists between RSL and glacial chronologies means that efforts to improve the quality of the existing RSL data base are an important research priority. Although isolation basins have been studied previously in Greenland for sea-level studies, their systematic analysis has been limited when compared with other areas in northwest Europe and North America. The data presented in this paper demonstrate the potential of such basins for developing high-resolution local RSL curves, from which intersite analyses of crustal rebound may proceed. The three main conclusions of this study are:

1. Isolation basins abound in West Greenland and provide a significantly under-utilised resource of high-resolution RSL data. A staircase of nine basins on Arveprinsen Ejland

in eastern Disko Bugt indicates a continuous fall in RSL from the marine limit at ca. 9.9 ka to 2.8 ka cal. yr BP (8.9 ka to 2.5 ka ^{14}C yr BP), at which time a switch from emergence to submergence took place. The RSL intersected present MSL at ca. 5 ka cal. yr BP (4.5 ka ^{14}C yr BP) and has risen by ca. 5 m since 2.8 ka cal. yr BP (2.5 ka ^{14}C yr BP).

2. A comparison with other RSL curves from Disko Bugt confirms previous suggestions of a west-east decrease in post-glacial rebound (e.g. Weidick, 1976; Funder, 1989; Rasch and Nielsen, 1994, 1995). Differences in RSL within east Disko Bugt are pronounced, however, and suggest that the orientation of the isobases here are not parallel to the Greenland Ice Sheet margin. Altitudinal differences in RSL between Arveprinsen Ejland and southeast Disko Bugt probably relate to the regional doming of the marine limit caused by enhanced ice loading and early deglaciation of the coastal zone south of the study area. This is perhaps not surprising when one considers the geometry of the coastal area. An additional conclusion from the above is that longitudinal data compilations, either in the form of shoreline relation or time-altitude graphs, should be avoided at least until the orientation of the isobases are better defined. This observation is a strong endorsement of the need for the development of further local RSL curves rather than the regional compilations typical of much previous research in Greenland.
3. The RSL on Arveprinsens Island has risen by ca. 5 m since ca. 2.8 ka cal. yr BP (2.5 ka ^{14}C yr BP). Comparisons with other data in the study area suggest that this late Holocene RSL rise began first in the eastern part of the bay, as might be expected under a scenario of crustal subsidence caused by neoglacial ice-sheet readvance (Kelly, 1980; Weidick, 1993). These trends in RSL are not compatible with those expected to accompany regional forebulge collapse, nor with recent geophysical model predictions of RSL change in the study area.

Acknowledgements We acknowledge the support and advice received from colleagues in the Danish Polar Center, particularly Dr Morten Rasch who helped in many ways and made valuable comments on a draft of this paper, as well as Associate Professor Niels Nielsen, Dr Ole Bennike and Dr Bent Fredskild. Logistical support was provided by the Arctic Station (University of Copenhagen), and we thank the Board of the Arctic Station and the crew of the *Porsild* for assisting us in transport within Disko Bugt and for advice regarding local conditions. Thanks are expressed to Mr Frank Lewis for the loan of My Way and to Shoprite Ltd, Total (Isle of Man), Alexander and Alexander, and Manx National Heritage. Much of this research was made possible by support received from the University of Durham Special Projects fund, which is gratefully acknowledged. The air photograph in Fig. 1 is reproduced with permission of Kort & Matrikelstyrelsen (A. 229–98). Diatom samples were prepared by Frank Davis and figures drawn by David Hume. Assistance in the preparation of field equipment by Ian Dennison, Derek Coates, Brian Priestly and Eddie Million is also gratefully acknowledged. Lastly, we appreciate the constructive reviews of this paper by Professor Alastair Dawson, Dr Ole Bennike and Dr Yongqiang Zong.

References

- BENNIKE, O. 1995. Palaeoecology of two lake basins from Disko, West Greenland. *Journal of Quaternary Science*, **10**, 149–155.
- BENNIKE, O., HANSEN, K. B., KNUDSEN, K. L., PENNEY, D. N. and RASMUSSEN, K. L. 1994. Quaternary marine stratigraphy and geochronology of central West Greenland. *Boreas*, **23**, 194–215.
- BJORCK, S., BENNIKE, O., POSSNERT, G., WOHLFARTH, B., and DIGERFELDT, G. 1998. A high-resolution ^{14}C dated sediment sequence from southwest Sweden: age comparisons between different components of the sediment. *Journal of Quaternary Science*, **13**, 85–89.
- CLARKE, J. A. and PRIMUS, J. A. 1987. Sea-level changes resulting from future retreat of ice sheets: an effect of CO_2 warming of the climate. *IN: Tooley, M. J. and Shennan, I. (eds), Sea-level Changes*, 356–370. Basil Blackwell, Oxford.
- CORNER, G. D. and HAUGANE, E. 1993. Marine-lacustrine stratigraphy of raised coastal basins and postglacial sea-level change at Lyngen and Vanna, Troms, northern Norway. *Norsk Geologisk Tidsskrift*, **73**, 175–197.
- DAHL-JENSEN, D., MOSEGAARD, K., GUNDESTRUP, N., CLOW, G. D., JOHNSEN, S. J., HANSEN, A. W. and BALLING, N. 1998. Past temperatures directly from the Greenland Ice Sheet. *Science*, **282**, 268–271.
- DANSGAARD, W., JOHNSEN, S. J., CLAUSEN, H. B., DAHL-JENSEN, D., GUNDESTRUP, N., HAMMER, C. U. and OESCHGER, H. 1984. North Atlantic climatic oscillations revealed by deep Greenland ice cores. *IN: Climate Processes and Climate Sensitivity*, 228–298. Proceedings of the Ewing Symposium, October 1982, Columbia University (Geophysical Monograph 29, Maurice Ewing Vol. 5). American Geophysical Union, Washington, DC.
- DONNER, J. 1978. Holocene history of the west coast of Disko, Central West Greenland. *Geografiska Annaler*, **60**, 63–72.
- DONNER, J. and JUNGER, H. 1975. Radiocarbon dating of shells from marine deposits in the Disko Bugt area, West Greenland. *Boreas*, **4**, 25–45.
- ELVERHØI, A., FJELDSKAAR, W., SOLHEIM, A., NYLAND-BERG, M. and RUSSWURM, L. 1993. The Barents Sea ice sheet – a model of its growth and decay during the last glacial maximum. *Quaternary Science Reviews*, **12**, 863–873.
- ESCHER, A., SØRENSEN, K. and ZECK, H. P. 1976. Nagssugtoqidian mobile belt in West Greenland. *IN: Escher, A. and Watt, S. (eds), Geology of Greenland*, 77–103. Geological Survey of Greenland, Denmark.
- FARVANDSVÆSENET 1995. *Tidevandstabeller Grønland 1996*. Forsvarsministeriets Tryk, Copenhagen, 111 pp.
- FOGED, N. 1972. The diatoms in four postglacial deposits in Greenland. *Meddelelser om Grønland*, **194**(4), 1–66.
- FOGED, N. 1973. Diatoms from Southwest Greenland. *Meddelelser om Grønland*, **194**(5), 1–84.
- FOGED, N. 1977. The diatoms in four postglacial deposits at Godthåbsfjord, West Greenland. *Meddelelser om Grønland*, **199**(4), 1–64.
- FREDSKILD, B. 1995. Palynology and sediment slumping in a high arctic Greenland lake. *Boreas*, **24**, 345–354.
- FREDSKILD, B. 1996. Holocene climate change in Greenland. *IN: Gronnow, B. (ed), The Paleo-Eskimo Cultures of Greenland. New Perspectives in Greenlandic Archaeology*, 243–251. Danish Polar Center, Copenhagen.
- FRICH, P. and INGOLFSSON, O. 1990. Det holocæne sedimentationsmiljø ved Igpiq samt en model for den relative landhævning i Disko Bugt, Vestgrønland. Dansk Geologisk Forening, Årsskrift for 1987–89, 1–10.
- FUNDER, S. 1989. Quaternary geology of the ice-free areas and adjacent shelves of Greenland. *IN: Fulton, R. J. (ed.), Quaternary Geology of Canada and Greenland*, 743–792. Geological Survey of Canada, Canadian Government Publishing Centre, Ottawa.
- FUNDER, S. and HANSEN, L. 1996. The Greenland ice sheet – a model for its culmination and decay during and after the last glacial maximum. *Bulletin of the Geological Society of Denmark*, **42**, 137–152.
- GREGORY, J. M. and OERLEMANS, J. 1998. Simulated future sea-level rise due to glacier melt based on regionally and seasonally resolved temperature changes. *Nature*, **391**, 474–476.
- HANSEN, B., HUMLUM, O., NIELSEN, N., RASCH, M. and FUNCH, P. 1997. Meteorological observations in 1996 at the Arctic Station, Qeqertarsuaq (Godhavn), Central West Greenland. *Geografisk Tidsskrift*, **97**, 152–153.

- HENDEY, N. I. 1964. *An Introductory Account of the Smaller Algae of British Coastal Waters. Part V: Bacillariophyceae (Diatoms)*. Fisheries Investigation Series, I, HMSO, London.
- HUSTEDT, F. 1957. Diatomeenflora der Fluss-Systeme der Weser im Gebiet der Hansestadt Bremen. *Abhandlungen Herausgegeben vom Naturwissenschaftlichen Verein zu Bremen*, **34**, 181–440.
- INGOLFSSON, O., FRICH, P., FUNDER, S. and HUMLUM, O. 1990. Palaeoclimatic implications of an early Holocene glacier advance on Disko Island, West Greenland. *Boreas*, **19**, 297–311.
- KALAND, P. E., KRZYWINSKI, K. and STABELL, B. 1984. Radiocarbon dating of transitions between marine and lacustrine sediments and their relation to the development of lakes. *Boreas*, **13**, 243–258.
- KELLY, M. 1980. The status of the Neoglacial in western Greenland. *Rapport Grønlands Geologiske Undersøgelse*, **96**, 1–24.
- KELLY, M. 1985. A review of the Quaternary geology of western Greenland. IN: Andrews, J. T. (ed.), *Quaternary Environments in Eastern Canadian Arctic, Baffin Bay and Western Greenland*, 461–501. Allen and Unwin, Boston.
- KJEMPERUD, A. 1986. Late Weichselian and Holocene shoreline displacement in the Trondheimsfjord area, central Norway. *Boreas*, **15**, 61–82.
- KRZYWINSKI, K. and STABELL, B. 1984. Late Weichselian sea level changes at Sotra, Hordaland, western Norway. *Boreas*, **13**, 159–202.
- NIELSEN, N., HANSEN, B. U., HUMLUM, O. and RASCH, M. 1995. Meteorological observations at Arctic Station, Qeqertasuaq (Godhavn), Central West Greenland. *Geografisk Tidsskrift*, **96**, 97–104.
- PALMER, A. J. M. and ABBOTT, W. H. 1986. Diatoms as indicators of sea-level change. IN: Van de Plassche, O. (ed.), *Sea-level Research: a Manual for the Collection and Evaluation of Data*, 457–488. Geo Books, Norwich.
- PATRICK, R. and REIMER, C. W. 1966. *The diatoms of the United States exclusive of Hawaii and Alaska*, Vol. 1. Monograph 13, The Academy of Natural Sciences of Philadelphia, Pennsylvania, 688 pp.
- PATRICK, R. and REIMER, C. W. 1975. *The diatoms of the United States exclusive of Hawaii and Alaska*, Vol. 2, Part 1. Monograph 13, The Academy of Natural Sciences of Philadelphia, Pennsylvania, 213 pp.
- PELTIER, W. R. 1996. Mantle viscosity and ice-age ice-sheet topography. *Science*, **273**, 1359–1364.
- RASCH, M. 1997. A compilation of radiocarbon dates from Disko Bugt, Central West Greenland. *Geografisk Tidsskrift*, **97**, 143–151.
- RASCH, M. In press. Holocene relative sea level changes in Disko Bugt, West Greenland. *Journal of Coastal Research*.
- RASCH, M. and JENSEN, J. F. 1997. Ancient Eskimo dwelling sites and Holocene relative sea-level changes in southern Disko Bugt, Central West Greenland. *Polar Research*, **16**, 101–115.
- RASCH, M. and NIELSEN, N. 1994. Holocene relative sea-level changes indicated by morphostratigraphic sequences; Sinigfik, Disko Island, West Greenland. *Geografisk Tidsskrift*, **94**, 37–45.
- RASCH, M. and NIELSEN, N. 1995. Coastal morpho-stratigraphy and Holocene relative sea level changes at Taupaat, southeastern Disko Island, central West Greenland. *Polar Research*, **14**, 277–289.
- RASCH, M., JAKOBSEN, B. H. and NIELSEN, N. 1997. Geomorphology and sedimentary record of three cusped forelands as indicators of late Holocene relative sea-level changes, Disko, West Greenland. *Geografisk Tidsskrift*, **97**, 33–46.
- RASMUSSEN, K. L. and RAHBK, U. 1996. The ^{14}C reservoir effect in Greenland. IN: Grønnow, B. (ed.), *The Palaeo-Eskimo Cultures of Greenland: New Perspectives in Greenlandic Archaeology*, 237–242. Danish Polar Center, Copenhagen.
- RETELLE, M. J., BRADLEY, R. S. and STUCKENRATH, R. 1989. Relative sea level chronology determined from raised marine sediments and coastal isolation basins, northern Ellesmere Island, Arctic Canada. *Arctic and Alpine Research*, **21**, 113–125.
- SHENNAN, I., INNES, J. B., LONG, A. J. and ZONG, Y. 1994. Late Devensian and Holocene relative sea-level changes at Loch nan Eala, near Arisaig, northwest Scotland. *Journal of Quaternary Science*, **9**, 261–283.
- SNYDER, J. A., FORMAN, S. L., MODE, W. N. and TARASOV, G. A. 1997. Postglacial relative sea-level history: sediment and diatom records of emerged coastal lakes, north-central Kola Peninsula, Russia. *Boreas*, **26**, 329–346.
- STABELL, B. 1985. The development and succession of taxa within the diatom genus *Fragilaria* Lyngbye as a response to basin isolation from the sea. *Boreas*, **14**, 273–286.
- STUIVER, M. and REIMER, P. J. 1993. Extended ^{14}C data base and revised CALIB 3.0 ^{14}C age calibration program. *Radiocarbon*, **35**, 215–230.
- TAUBER, H. 1960a. Copenhagen natural radiocarbon measurements III, corrections to radiocarbon dates made with the solid radiocarbon technique. *American Journal of Science Radiocarbon Supplement*, **2**, 5–11.
- TAUBER, H. 1960b. Copenhagen radiocarbon dates IV. *American Journal of Science Radiocarbon Supplement*, **2**, 12–25.
- TAUBER, H. 1964. Copenhagen radiocarbon dates VI. *Radiocarbon*, **6**, 215–225.
- TROELS-SMITH, J. 1955. Characterisation of unconsolidated sediments. *Danmarks Geologiske Undersøgelse, Series IV*, **3**, 38–73.
- VAN DER WERFF, H. and HULS, H. 1958–1974. *Diatomeenflora van Nederland*, 8 parts. Published privately: De Hoef, The Netherlands.
- WEIDICK, A. 1972. Holocene shorelines and glacial stages in Greenland — an attempt at correlation. *Rapport Grønlands Geologiske Undersøgelse*, **41**, 1–39.
- WEIDICK, A. 1976. Glaciation and the Quaternary. IN: Escher, A. and Watt, S. (eds), *Geology of Greenland*, 431–458. Geological Survey of Greenland, Copenhagen.
- WEIDICK, A. 1985. Review of glacier changes in West Greenland. *Zeitschrift für Gletscherkunde und Glazialgeologie*, **21**, 301–309.
- WEIDICK, A. 1992. Landhævning og landsenkning i Grønland siden sidste istid. *Naturens Verden*, **3**, 81–96.
- WEIDICK, A. 1993. Neoglacial change of ice cover and the related response of the Earth's crust in West Greenland. *Rapport Grønlands Geologiske Undersøgelse*, **159**, 121–126.
- WEIDICK, A. 1996. Neoglacial changes of ice cover and sea level in Greenland — a classical enigma. IN: Grønnow, B. (ed.), *The Palaeo-Eskimo Cultures of Greenland: New Perspectives in Greenlandic Archaeology*, 257–270. Danish Polar Center, Copenhagen.
- WEIDICK, A., OERTER, H., REEH, N., THOMSEN, H. H. and THORNING, L. 1990. The recession of the Inland Ice margin during the Holocene climatic optimum in the Jakobshavn Isfjord area of West Greenland. *Palaeogeography, Palaeoclimatology and Palaeoecology (Global Planetary Change Section)*, **82**, 389–399.